THE EVOLUTION OF MACHINE LEARNING AND AI

Week 1 Reference Booklet

The Shape of the Field: Cycles, Breakthroughs, and What the Current Wave Got Right

Award	Certificate in Artificial Intelligence and Machine Learning Engineering (30 ECTS)
Parent programme	Higher Diploma in Science in Computer Science (NFQ Level 8)
Module	The Evolution of Machine Learning and AI (5 ECTS, Semester 1)
Week	Week 1 of 12 · Sprint 1: Fast foundations and responsible AI up front
Indicative content	Item 1 (History of AI/ML: early pioneers, AI winters, resurgence) and Item 5 (Influential breakthroughs)
Learning outcome	LO1: Trace the historical development of AI and machine learning
Assessed in	Mid-module quiz, Week 6 (50% of module mark)
Companions	Week 1 Interactive Companion (hype-wave timeline) · Week 1 live lecture · Lab: environment setup

How to read this booklet. The live lecture gives you the story in two hours; this booklet gives you the full account. Everything in Chapters 1 to 13 is examinable in the Week 6 quiz. Boxes marked **Deeper Detail** go beyond what the lecture had time for. Boxes marked **Everyday Analogy** are the mental pictures the lecture used: they are not decorations, they are the fastest route to remembering the concept under exam conditions.

Contents

1 Why This History Matters

COVERS	Orientation: what this week is for, and the one pattern to watch
TIME	Read first, about 15 minutes
IN BRIEF	AI has crashed twice. The crashes follow a repeatable pattern, and knowing that pattern is a professional skill, not trivia. This chapter explains what you will be able to do by the end of the booklet.

You could learn to train a neural network without knowing any of the history in this booklet. Plenty of working engineers have. So it is fair to ask, before you commit to fifty pages: why does a hands-on engineering module begin with seventy-five years of stories?

The answer is that artificial intelligence is not a field with a smooth past. It is a field that has **crashed twice**, publicly and expensively, and the forces that caused those crashes are active right now, in the products you will build and the news you will read this week. The history is not background. It is a user manual for the present.

1.1 Three practical skills this week gives you

First: you will read AI news critically. Somewhere this month, a headline has claimed that a new system “reasons like a human” or is “on the verge of general intelligence”. By the end of this booklet you will know that a nearly identical claim was printed about a machine in 1958, that the machine in question could see four hundred pixels, and what happened to the field over the following fifteen years as a result. That knowledge changes how you read the headline. It does not make you a cynic; the 1958 machine really was the ancestor of everything we now use. It makes you **calibrated**, which is different, and rarer, and worth more.

Second: you will make better engineering decisions. The 1980s expert systems boom is, on the surface, a story about old software. Underneath, it is the clearest lesson ever taught about **when hand-written rules are the wrong tool and learning from data is the right one**. That is a judgement call you will personally face in industry, probably within a year of graduating: a stakeholder will ask whether some behaviour should be coded as rules or learned from examples, and the correct answer will depend on exactly the factors that killed the expert systems industry. History here is compressed engineering experience, paid for by other people.

Third: you will understand the module thesis. This module argues one big thing across twelve weeks: the current AI wave succeeded through brute force (more data, more computing power, bigger models), that recipe is now hitting real walls, and the frontier is turning toward two responses, **grounding models in science** (Week 5) and **making models lean** (Weeks 8 and 9). You cannot evaluate that thesis without knowing what the brute-force recipe was, why it worked when nothing before it did, and what “walls” have looked like in the past. That is this week’s job.

1.2 The one picture to keep in your head

If you retain a single image from Week 1, make it this one: the field’s history is a **wave**. Enthusiasm and funding rise on a genuine breakthrough, overshoot into impossible promises, collapse when reality arrives, then quietly rebuild under a different name until the next breakthrough. The two collapses are famous enough to have a name: the **AI winters**.

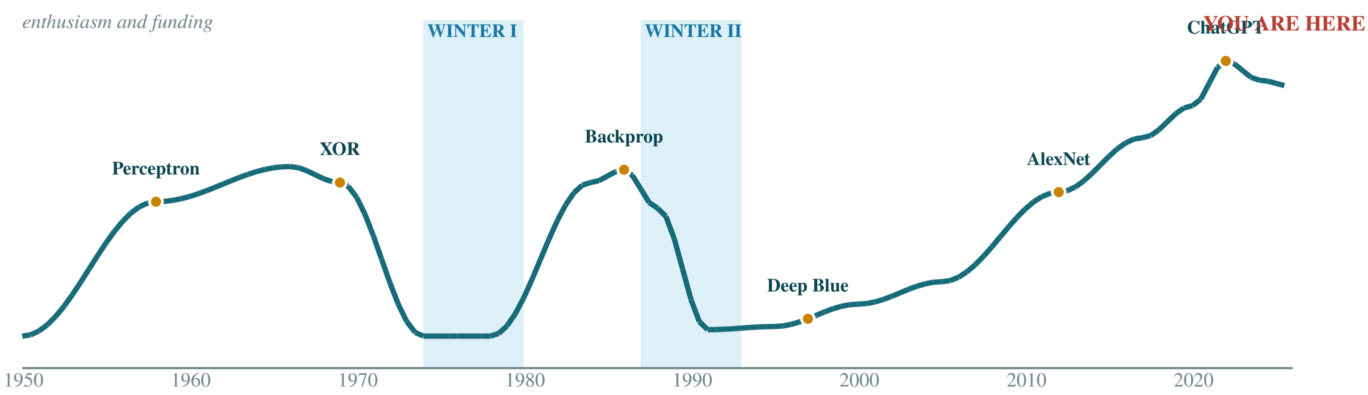

Figure 1.1 · Seventy-five years of enthusiasm and funding. The two shaded bands are the winters. The underlying science advanced straight through both.

KEY IDEA

The AI cycle: breakthrough → hype → promises outrun results → funding collapse (a "winter") → quiet progress under other names → next breakthrough. It has completed two full turns (winters in the mid 1970s and late 1980s) and we are currently high on the third rise. Whether this rise ends in a third winter, a plateau, or something genuinely different is an open question, and by Week 12 you should have your own defensible answer.

EVERYDAY ANALOGY

Think of a seaside town's economy. A genuinely good beach (the breakthrough) attracts visitors. Speculators build far more hotels than the beach can fill (the hype). A bad season exposes the overbuilding, hotels go bankrupt, and the town gets a reputation as a failure (the winter), even though the beach was real all along. Years later, someone reopens a hotel at a sensible size and the cycle begins again. The beach never stopped being good. The **promises about the beach** were the problem. In AI, the underlying science kept advancing straight through both winters; it was the promises, and the funding attached to them, that froze.

1.3 How the booklet is organised

Chapters 2 to 13 tell the story in order: the prehistory of the idea, Turing's founding question, the naming of the field at Dartmouth, the first boom and the first winter, the expert systems boom and the second winter, the quiet statistical revolution, the 2012 deep learning breakthrough, the transformer and the scaling era, and the walls the field is hitting right now, which is where the module thesis takes over.

Chapter 14 is a field guide to reading AI news, applying everything before it. Chapters 15 to 17 are your revision core: the whole pattern compressed to two pages, a master timeline, a plain-English glossary, and a bank of practice questions in the style of the Week 6 quiz. Chapter 18 collects further reading and viewing, all optional.

Two reading routes are sensible. The **full route** is front to back, ideally within a week of the lecture, at roughly two hours. The **revision route**, for the days before the Week 6 quiz, is Chapter 15 first, then the Quiz Watch boxes scattered through the booklet, then the question bank in Chapter 17, dipping back into full chapters only where a question exposes a gap.

QUIZ WATCH · WEEK 6

The Week 6 quiz tests LO1 partly through **pattern questions**: given a description of some episode (real or invented), identify which stage of the AI cycle it represents and justify the identification in two or three sentences. The seaside town analogy above is a reliable scaffold for such answers.

2 The Prehistory: Dreams, Logic, and a Dangerous Idea

COVERS	From ancient myths to the mathematics that made AI conceivable, roughly 1650 to 1949
TIME	About 20 minutes
IN BRIEF	The idea of an artificial mind is ancient; the mathematics that made it buildable arrived between the 1840s and the 1940s. Four names carry the chapter: Lovelace, Boole, Turing (1936), and McCulloch and Pitts.

Artificial intelligence did not begin with computers. The dream of built minds is one of humanity's oldest stories, and the mathematics that turned the dream into an engineering project was in place before a single electronic computer existed. This chapter is short on machinery and long on ideas, because the ideas are what everything later stands on.

2.1 The dream is ancient

Greek myth gave us **Talos**, a bronze giant built by the smith-god Hephaestus to patrol the shores of Crete, arguably literature's first security robot. Jewish folklore gave us the **golem**, a clay servant animated by sacred words, complete with the genre's first alignment failure: golem stories routinely end with the creation following instructions too literally and causing disaster. Mary Shelley's **Frankenstein** (1818) carried the theme into the scientific age and named its central anxiety: responsibility for what we build. When Week 2 introduces accountability as an engineering requirement, it is joining a conversation that is at least two thousand years old.

Alongside the stories ran a quieter tradition of **automata**: mechanical ducks, chess-playing figures (the famous eighteenth-century "Mechanical Turk" was a hoax with a human hidden inside, a name Amazon later borrowed with a wink for its human-powered task service), and elaborate clockwork figures that wrote or played instruments. The automata mattered less for what they did than for the question they normalised: **how much of what we call intelligence is mechanism?**

2.2 The mathematics assembles, piece by piece

2.2.1 Leibniz's dream and Boole's laws

The philosopher and mathematician Gottfried Leibniz (1646-1716) imagined a "calculus of reasoning": a symbolic language in which disagreements could be settled by computation, with disputants saying, in his words, **let us calculate**. It was a fantasy in his lifetime, but it planted the field's founding assumption: that reasoning itself might be a kind of arithmetic.

In 1854, the English mathematician George Boole published **An Investigation of the Laws of Thought**, showing that logical reasoning (AND, OR, NOT) could be written as algebra over the values true and false. Boolean algebra looked like a philosophical curiosity for ninety years, until a masters student named Claude Shannon showed in 1937 that it exactly describes what electrical switching circuits do. That single observation is why computers can reason about anything at all: logic had become wiring. Remember Shannon's name; he reappears at the founding of AI itself in Chapter 4.

2.2.2 Babbage, Lovelace, and the first debate about machine creativity

In the 1830s and 1840s, Charles Babbage designed (though never completed) the **Analytical Engine**, a general-purpose, programmable mechanical computer, a century early. His collaborator **Ada Lovelace** wrote what is generally regarded as the first published algorithm for it, and, more importantly for us, the first considered argument about the limits of machine intelligence. In her 1843 notes she wrote that the Engine “has no pretensions whatever to **originate** anything. It can do whatever we know how to order it to perform.”

That sentence, now called **Lovelace’s objection**, is the oldest move in the AI debate, and you will hear it this week from friends and family in its modern form: “AI just remixes what it was trained on; it cannot create anything new.” Whether the objection survives contact with modern systems is genuinely contested, and Turing himself answered it directly in 1950, as Chapter 3 describes. What matters here is the shape of the argument: from the very first program, humans have disputed whether machines can go beyond their instructions. Machine **learning**, the subject of this entire module, is precisely the technology that complicates Lovelace’s claim, because a system trained on examples was never explicitly “ordered” to do most of what it ends up doing.

DEEPER DETAIL: Hilbert’s programme and the strange birth of computer science

Here is one of history’s better ironies: computer science was born from a **failed** attempt to prove computers unnecessary. Around 1900, the great mathematician David Hilbert challenged the field to show that all of mathematics could be derived mechanically from axioms, and that a definite procedure could decide the truth of any mathematical statement (the **Entscheidungsproblem**, or “decision problem”). If Hilbert was right, mathematics would in principle be automatable, and mathematicians reducible to machine operators.

Two young logicians demolished the programme. Kurt Gödel proved in 1931 that any consistent formal system rich enough for arithmetic contains true statements it cannot prove (the incompleteness theorems). Then in 1936 Alan Turing, aged 23, attacked the decision problem, and to do it rigorously he first had to **define** what “a definite procedure” means. His definition was a thought experiment: an idealised machine reading and writing symbols on an infinite tape according to a finite table of rules, now called the **Turing machine**.

Turing proved two things. First, no such machine can solve the decision problem: some questions are undecidable, so Hilbert’s dream fails. Second, and this is the part that built the modern world, a single **universal** machine can simulate any other machine, given that machine’s description as input. In plain terms: one sufficiently flexible device can run **any** program. That is the theoretical blueprint of the general-purpose computer, published before any existed, discovered as a side effect of proving a point in pure logic. Every laptop, phone, and GPU cluster is a physical approximation of Turing’s 1936 thought experiment.

2.2.3 McCulloch and Pitts: the brain as logic (1943)

The final piece of prehistory belongs to an unlikely pair: Warren McCulloch, a middle-aged neurophysiologist, and Walter Pitts, a homeless teenage logic prodigy who had taught himself mathematics in public libraries and once sent corrections to Bertrand Russell. Their 1943 paper, **A Logical Calculus of the Ideas Immanent in Nervous Activity**, proposed a radically simplified model of the brain cell: a unit that sums its inputs and “fires” if the total crosses a threshold.

Their headline result: networks of these simplified neurons can implement any logical function. Combined with Boole (logic is algebra), Shannon (algebra is circuitry), and Turing (one machine can compute anything computable), the syllogism practically wrote itself: **if brains compute, then machines can, in principle, do what brains do.** Every neural network in this module, up to and including the largest language models on Earth, is a descendant of the McCulloch-Pitts neuron, with one crucial addition that Chapter 5 introduces: the ability to **learn** its own settings rather than having them fixed by hand.

KEY IDEA

By 1949, before the field had a name, its intellectual foundation was complete: reasoning can be algebra (Boole), algebra can be circuitry (Shannon), one machine can run any program (Turing 1936), and simplified brain cells can implement logic (McCulloch and Pitts). Artificial intelligence begins not as an invention but as an **inference** from these four results.

The cyberneticians: the almost-founders

In the 1940s a glamorous interdisciplinary movement called **cybernetics**, led by the mathematician Norbert Wiener, studied feedback and control in animals and machines: how a thermostat, a steersman, and a hand reaching for a cup all correct themselves toward a goal. Cybernetics conferences brought neuroscientists, engineers, and anthropologists to the same table, and for a while it looked like the science of intelligent machinery would be born there. It was not, partly for reasons of personality and politics: when John McCarthy chose a name for the 1956 Dartmouth workshop (Chapter 4), he later admitted that one attraction of the fresh phrase "artificial intelligence" was precisely that it avoided cybernetics and the obligation to argue with Wiener. Fields, like products, are sometimes shaped by branding decisions. Cybernetics' core ideas, feedback and goal-correction, quietly re-entered AI decades later inside reinforcement learning, which reappears in Week 7 as the technique that turned raw language models into usable assistants.

QUIZ WATCH · WEEK 6

A favourite short-answer pattern: **"Explain Lovelace's objection and one modern response to it."** A strong answer states the objection (machines only do what they are ordered to do, so they cannot originate), then notes that machine learning complicates it, because learned behaviour is not explicitly ordered, and that Turing's 1950 reply (Chapter 3) argued machines regularly surprise their own designers.

3 Turing's Question: 1950

COVERS	Alan Turing's 1950 paper, the Imitation Game, his predictions, and his replies to nine objections
TIME	About 20 minutes
IN BRIEF	Turing converted "can machines think?" from philosophy into an engineering target: judge the machine by its conversational behaviour. His paper anticipated most of today's AI debate, seventy-five years early.

In October 1950, the journal **Mind** published a paper by Alan Turing titled **Computing Machinery and Intelligence**. It is one of the few genuinely foundational documents in any field that is also a pleasure to read: witty, generous to opponents, and shorter than most modern survey papers. If you read one primary source this semester, make it this one.

3.1 Throwing away the question

The paper opens with the obvious question, "Can machines think?", and then does something surprising: it refuses to answer, on the grounds that the words "machine" and "think" are too vague for the question to mean anything. Arguing about definitions, Turing suggests, is a way of generating heat without light. So he replaces the question with a **test**.

His **Imitation Game**, now universally called the **Turing Test**, works like this. A human judge holds two text conversations, one with a hidden human and one with a hidden machine, and may ask anything. If, after sustained questioning, the judge cannot reliably tell which is which, the machine passes. No claims about consciousness, no peering inside; just behaviour, measured.

EVERYDAY ANALOGY

Judge the cook by the food, not by inspecting their soul. Turing's move is to say: we never verify that **other people** think by examining their inner life either; we infer it from how they behave, chiefly from how they converse. Fairness demands we offer machines the same court and the same evidence. Whether passing the test constitutes "real" thinking is a question Turing deliberately leaves at the door, the way a chef leaves philosophy out of the kitchen.

The move is profoundly practical, and it set the culture of the field. AI became, from birth, a discipline that advances by **benchmarks**: shared, measurable targets that anyone can attempt and everyone can score. The ImageNet contest that launches the modern era in Chapter 10 is a direct cultural descendant of the Imitation Game, and so is every leaderboard the field argues about today.

3.2 The predictions, and how they aged

Turing made concrete forecasts, which is itself remarkable: most philosophy avoids falsifiable claims. He predicted that by about the year 2000, machines with roughly a billion units of storage would fool an average questioner 30 percent of the time after five minutes of conversation, and that by then "one will be able to speak of machines thinking without expecting to be contradicted."

Score him honestly and he comes out startlingly well. The storage estimate was in a defensible range for the era he named. The conversational bar was reached late, roughly two decades after his date, but decisively: modern chat systems clear a five-minute casual test with ease, and the interesting arguments have moved to longer, adversarial evaluations. And his sociological prediction, that educated opinion would drift toward speaking of machine thinking unselfconsciously, has simply come true; listen to how colleagues talk about the tools they use daily. Being twenty years late on a fifty-year forecast about an unborn technology is elite forecasting.

DEEPER DETAIL: The nine objections, and why they still run the debate

The heart of the paper is Turing's list of nine objections to machine intelligence, each stated fairly and answered. Three matter most for this module.

The argument from consciousness says a machine cannot think unless it **feels**: unless there is something it is like to be the machine. Turing's reply is that this standard, applied consistently, collapses into solipsism: you cannot verify anyone's inner experience but your own, yet you politely grant that other people think. He suggests we extend machines the same courtesy once their behaviour warrants it. Seventy-five years on, this is still exactly where the consciousness debate stands.

Lady Lovelace's objection (Chapter 2) says machines only do what they are ordered to do, so they cannot originate anything. Turing's reply is disarmingly empirical: "machines take me by surprise with great frequency." He notes that surprise usually reveals the limits of the **programmer's** foresight rather than any creative spark, but presses the point that the objection quietly assumes all consequences of an instruction are foreseen by whoever gives it, which is false for any interesting program. Machine learning sharpens his reply enormously: a system trained on examples was never explicitly ordered to do most of what it does, and modern systems produce moves and strategies their builders demonstrably did not anticipate, as the AlphaGo story in Chapter 11 shows in the most public way imaginable.

The mathematical objection leans on Gödel and Turing's own 1936 result: since formal systems have unprovable truths, machines have in-principle limits, so minds must be more than machines. Turing's reply: the argument assumes, without evidence, that humans are **not** subject to comparable limits. We feel unlimited from the inside, but so might any sufficiently complex machine.

The remaining objections (theological, "heads in the sand", arguments from disabilities like humour or falling in love, from continuity of the nervous system, from informality of behaviour, and from extrasensory perception, which was taken more seriously in 1950 than you might expect) are worth reading in the original. The practical takeaway: nearly every argument you will hear about AI at a dinner table in 2026 appears in this 1950 paper, usually in a stronger form than the dinner-table version, alongside a courteous rebuttal.

The man himself

Turing's biography is essential context for how the field remembers him. During the Second World War he was central to the codebreaking effort at Bletchley Park that broke German naval Enigma, work of decisive strategic value that remained secret for decades, meaning the founder of computer science was, in his lifetime, publicly credited with almost none of his most consequential work. In 1952 he was prosecuted by the British state for homosexuality, then a crime, stripped of his security clearance, and sentenced to chemical castration. He died in 1954, aged 41, of cyanide poisoning, ruled a suicide. He never saw the field his paper founded hold its first workshop. The British government formally apologised in 2009; a royal pardon followed in 2013; his face now appears on the fifty-pound note. The field's highest honour, the Turing Award, which recurs throughout this booklet, carries his name.

QUIZ WATCH · WEEK 6

Be able to do three things with this chapter: state what the Turing Test replaces and why (vague definitional questions with a measurable behavioural test); name and explain one of Turing's objections **and** his reply in your own words; and connect the test to the field's benchmark culture in one sentence.

4 The Christening: Dartmouth, 1956

COVERS The workshop that named the field, the founding generation, and the two rival families of AI
TIME About 15 minutes
IN BRIEF A summer workshop at Dartmouth College gave the field its name, its founding personalities, and its first overdose of optimism. Two rival philosophies, rules versus learning, emerge here and alternate in dominance for the next seventy years.

In the summer of 1956, a 28-year-old mathematician named John McCarthy convened a workshop at Dartmouth College in New Hampshire. The funding proposal he co-wrote with Marvin Minsky, Nathaniel Rochester, and Claude Shannon is the field's birth certificate, and two sentences in it did most of the work.

The first coined the name: the proposal was for a "2 month, 10 man study of **artificial intelligence**". McCarthy chose the phrase deliberately: it was vivid, it was unclaimed, and, as he later admitted, it side-stepped the existing field of cybernetics and its dominant personality, Norbert Wiener (Chapter 2). A field's name is a strategic act, a lesson the field would relearn in reverse during the second winter, when "artificial intelligence" became so toxic that researchers rebranded their work as "machine learning" to survive (Chapter 8).

The second sentence set the tone for everything that followed: "every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it." Note what this is: not a finding, but a **conjecture**, stated as a premise. The proposal went on to suggest that a significant advance could be made in a single summer by a carefully selected group. That estimate was off by, so far, seventy years, and counting.

4.1 The founding generation

The attendees and near-attendees of Dartmouth ran the field for the next three decades. **John McCarthy** went on to invent the LISP programming language (for decades the native tongue of AI research, and the reason a hardware industry in Chapter 7 exists) and founded the Stanford AI Lab. **Marvin Minsky** co-founded the MIT AI Lab and became the field's most quotable optimist and, in 1969, the author of its most consequential critique (Chapter 6). **Claude Shannon**, already famous as the father of information theory, lent the enterprise credibility. **Allen Newell** and **Herbert Simon** arrived from Carnegie Tech with the only working system at the workshop: the Logic Theorist, a program that proved theorems from **Principia Mathematica**, in one case finding a proof more elegant than the published one. Simon, who would later win a Nobel Prize in economics, was the group's boldest forecaster, and his forecasts are worth recording precisely because of how instructively they failed.

The forecasts of 1957 to 1965

Simon and Newell, 1957: within ten years a computer will be world chess champion, and will discover and prove an important new mathematical theorem. (Chess took forty years, Chapter 9; the theorem is still pending.) Simon, 1965: "machines will be capable, within twenty years, of doing any work a man can do." Minsky, 1967: "within a generation, the problem of creating artificial intelligence will substantially be solved," and, reportedly, in 1970: three to eight years until a machine with the general intelligence of an average human being.

These were not fools; they were among the most brilliant scientists of their century, speaking from genuine early success. That is exactly why the pattern matters. The lesson is not "pioneers were silly" but "**early success on toy versions of a problem is weak evidence about the full problem**", because difficulty in AI scales in ways that are profoundly unintuitive. Hold on to this when Chapter 5 introduces microworlds and

Chapter 13 discusses today's forecasts about general intelligence; the epistemics of 1965 and 2026 rhyme uncomfortably well.

4.2 The two rival families

Dartmouth also crystallised a split that organises the entire history of the field, and this booklet. Two philosophies of how to build intelligence emerged, and they alternate in dominance from here to the present day.

Symbolic AI, later nicknamed "good old-fashioned AI", holds that intelligence is the manipulation of symbols according to logical rules: facts in, rules applied, conclusions out, like algebra over knowledge. If you can write the rules down, the machine can be intelligent. Its natural products are theorem provers, planners, and the expert systems of Chapter 7. Its natural strength is **transparency**: you can read the rules and audit the reasoning. Its natural weakness will emerge painfully across the next three chapters.

Connectionism holds that intelligence **emerges** from many simple, brain-inspired units connected in networks that **learn** from experience rather than being told rules. Its natural product is the neural network, from Chapter 5′s perceptron to the largest language models. Its natural strength is learning what no one can write down; its natural weakness, opacity, is a running theme of Week 2′s work on explainability.

KEY IDEA

Keep the two families in view at all times: **rules written by hand** versus **behaviour learned from data**. Symbolic AI dominated from 1956 into the 1990s and crashed twice; connectionism spent thirty years as the losing side, then won nearly everything after 2012. The most interesting current work, previewed in Chapter 13, quietly recombines them: science-grounded models build known equations (symbolic knowledge) into learning systems, and modern reasoning systems wrap learned models in structured search. The pendulum may be becoming a synthesis, and you are studying the field at the moment it happens.

QUIZ WATCH · WEEK 6

Reliable quiz territory: what the Dartmouth proposal claimed and why the claim mattered (a conjecture stated as a premise, setting the field's optimism); one founding figure and their contribution; and a two-sentence contrast of symbolic AI and connectionism with one strength and one weakness each.

5 The Golden Years and the First Learning Machine: 1956 to 1969

COVERS	Early triumphs: reasoning programs, LISP, microworlds, ELIZA, Shakey, and Rosenblatt's perceptron
TIME	About 25 minutes
IN BRIEF	For a dozen years everything seemed to work. Programs proved theorems, solved puzzles, and, with the perceptron, **learned**. This chapter explains the learning loop that still trains every modern model, and the illusions that set up the crash.

The dozen years after Dartmouth felt, to those inside, like watching prophecy fulfil itself. Money flowed (chiefly from the US Department of Defense through ARPA, which funded people rather than projects and asked few questions), the founding labs at MIT, Stanford, and Carnegie Mellon grew fat, and a parade of programs did things machines had never done.

5.1 What the golden years actually built

Newell and Simon followed the Logic Theorist with the **General Problem Solver**, whose core technique, means-ends analysis, is worth knowing: compare the current state to the goal, find the biggest difference, and apply an operation that shrinks it, recursing on any obstacles. It is, recognisably, how you plan a house move. The name reveals the era's ambition: not **a** problem solver but the **general** one; in practice it handled well-formalised puzzles and little else, which was the era in miniature.

McCarthy invented **LISP** (1958), a programming language built around symbols and lists, so suited to AI work that it remained the field's standard for over thirty years and eventually spawned a dedicated hardware industry, whose fate is a load-bearing plot point in Chapter 8. Danny Bobrow's STUDENT solved algebra word problems; Tom Evans' ANALOGY solved the geometric-analogy items from IQ tests; at Stanford, **Shakey the robot** combined vision, planning, and movement for the first time, and earned a Life magazine profile calling it the "first electronic person", a phrase to remember next time you read a 2026 headline.

The characteristic method of the era was the **microworld**: a radically simplified domain in which the messy real world could not intrude. The most famous was the blocks world, a tabletop of coloured blocks to stack and describe, and its most famous inhabitant was Terry Winograd's SHRDLU (1970), which held genuinely impressive dialogues about its blocks: it could answer "can a pyramid support a block?", execute "put the blue block on the red one", and even justify its own past actions. Within the blocks world, SHRDLU **understood**.

> **EVERYDAY ANALOGY**
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> A microworld is a swimming pool with the shallow end roped off. Learning to float there is real progress, and everything you learn is true. The danger is inferring readiness for the open ocean, which differs from the pool not in degree but in kind: currents, waves, cold, and no walls. The golden-years programs were superb in the roped-off shallow end, and their builders, in good faith, extrapolated to the ocean. Chapter 6 is about what the ocean did to those extrapolations.

5.2 The perceptron: the machine that learns

The most important artefact of the era, for this module, came from outside the symbolic mainstream. In 1958, Frank Rosenblatt, a psychologist at the Cornell Aeronautical Laboratory, demonstrated the **perceptron**: a McCulloch-Pitts style artificial neuron with the crucial addition the 1943 model lacked. Its settings were not fixed by hand. **It adjusted them itself, from examples.** This is the ancestral form of everything you will train in this course, so it is worth slowing down for.

5.2.1 The learning loop, in plain language

Imagine the perceptron deciding whether a small image shows a cat.

1. **Weighted evidence.** Each input (each pixel's brightness) is multiplied by a **weight**: a dial that says how strongly, and in which direction, this input should push the verdict. Weights can be positive (evidence for), negative (evidence against), or near zero (irrelevant).
2. **Sum and threshold.** Add up all the weighted evidence. Over the threshold, output "cat"; under it, "not cat". So far this is just McCulloch and Pitts.
3. **The magic step: learn from error.** Show it a labelled example. If the verdict is **correct, change nothing**. If it is **wrong**, nudge every dial a small step in the direction that would have made it less wrong: inputs that should have pushed toward the right answer get a little more weight, the rest a little less.
4. **Repeat** over thousands of examples. The dials settle into a configuration that separates cats from non-cats, and, remarkably, nobody ever told the machine what a cat looks like. The knowledge is **in the dials**, put there by feedback alone.

> **EVERYDAY ANALOGY**
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> Learning darts by throwing. You do not study the physics of projectiles; you throw, see where the dart lands, and adjust your arm slightly toward the target. Guess, check, adjust, repeat, until your arm "knows" something your conscious mind could never write down as rules. That three-beat loop, guess, check, adjust, wrapped in a century of mathematics, is **still how the largest models on Earth are trained**. When Week 3 introduces gradient descent and backpropagation, recognise them as this loop made precise: gradient descent is the rule for which direction to nudge the dials, and backpropagation is the bookkeeping that assigns each of billions of dials its share of the blame for an error.

Rosenblatt also proved a genuine theorem, the **perceptron convergence theorem**: if a set of examples **can** be separated by his machine, the learning procedure is guaranteed to find a separating configuration in finite time. A learning machine with a mathematical guarantee, in 1958, is rightly regarded as a landmark. The fine print, "if it can be separated", turns out to contain the next chapter.

> ***Four hundred pixels and a front page***
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> The hardware was the Mark I Perceptron: a camera feeding a 20 by 20 grid of photocells, four hundred pixels, wired through a patch panel to motor-driven potentiometers, physical dials turned by motors, that held the weights. Learning was literally audible: the machine whirred as it adjusted its opinion. It could learn to distinguish simple shapes and letters.
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> Then came the press conference. In July 1958, following a Navy demonstration, the **New York Times** reported that the perceptron was "the embryo of an electronic computer that the Navy expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence." Rosenblatt, charismatic and quotable, did little to dampen such coverage, and rival researchers, particularly in the symbolic camp, seethed at what they saw as showmanship.
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> Hold the two facts side by side: a four-hundred-pixel shape classifier, and a national newspaper reporting a forthcoming conscious machine. That gap, between a real, genuinely important result and the promises stapled to it, is the purest specimen of AI hype ever collected, and it was printed two years into the field's existence. The rivalry it inflamed, between Rosenblatt's connectionists and the symbolic establishment, is the fuse that Chapter 6 lights.

5.3 ELIZA and the fluency trap: 1966

One more golden-years artefact matters too much to compress. In 1966, Joseph Weizenbaum at MIT built **ELIZA**, a conversation program running a script called DOCTOR that imitated a Rogerian psychotherapist, a style of therapy that consists largely of reflecting the patient's words back as questions. ELIZA worked by pattern-matching: find a keyword, apply a template. "I am sad about my job" becomes "Why do you say you are sad about your job?" It understood nothing whatsoever, and its author knew it; he had built it partly to **demonstrate** the superficiality of such tricks.

The demonstration backfired spectacularly. Users confided in ELIZA at length. Weizenbaum's own secretary, who had watched him build it and knew exactly what it was, asked him to leave the room so she could talk to it privately. Some clinicians speculated in print about automated therapy. Weizenbaum was so disturbed that he spent much of his remaining career as the field's in-house moral critic, arguing in his 1976 book **Computer Power and Human Reason** that some things machines **could** do, they **should not** do, an argument Week 2 picks up in modern dress.

KEY IDEA

The **ELIZA effect**: humans reflexively project understanding, and even care, onto anything that produces fluent, contextually appropriate language. It is a fact about **us**, not about the machine, which is why it did not fade as machines improved; it compounded. ELIZA ran on a few hundred templates. Modern systems are fluent beyond any 1966 imagining, and the projection they invite is proportionally stronger. Carry the ELIZA effect into Week 2 (it is why transparency about what a system is matters ethically) and into Week 7 (it is the standing reason to test language models on substance, never on fluency, because fluency is precisely the thing they cannot help but have).

TRY IT YOURSELF

ELIZA clones run in any browser; the lab links one. Spend five minutes talking to it, and notice two things in yourself: how quickly the illusion forms when you play along, and what single question most brutally collapses it. Write that question down. Then, this week, put a structurally similar question to a modern chatbot and observe what has and has not changed in sixty years. This makes an excellent forum post.

QUIZ WATCH · WEEK 6

From this chapter, be able to: walk through the perceptron's learning loop in four plain-English steps and connect it by name to gradient descent; state what the convergence theorem guarantees **and** its fine print; define the microworld method and its risk using the swimming-pool analogy; and define the ELIZA effect with its Week 2 relevance in one sentence.

6 XOR and the First Winter: 1969 to 1980

COVERS The Minsky-Papert critique, the XOR problem worked in full, the Lighthill Report, and the anatomy of a funding collapse

TIME About 25 minutes

IN BRIEF A rigorous, correct piece of mathematics about the **simplest** neural network was read as a verdict on the **whole idea**, and a decade of overpromising ran out of road at the same moment. The result was the first AI winter.

Every winter needs two ingredients: an intellectual trigger and an economic one. The first winter had a textbook example of each.

6.1 The intellectual trigger: Perceptrons, 1969

In 1969 Marvin Minsky and Seymour Papert, both of MIT's symbolic camp, published **Perceptrons**, a mathematically careful book analysing exactly what Rosenblatt's single-layer machine could and could not learn. Its most famous result concerns a logical function so simple you use it daily without noticing.

6.1.1 XOR, worked in full

XOR, "exclusive or", is true when exactly one of two inputs is true: one or the other, but not both.

EVERYDAY ANALOGY

A stairwell light wired to two switches, one at the bottom of the stairs and one at the top. Flipping **either** switch changes the light. Work out the pattern: if both switches are down, the light is off; flip one (either one) and it is on; flip the second and it is off again. The light is ON exactly when **one** switch is up, OFF when zero or both are. That is XOR, implemented in the wiring of nearly every two-storey building you have entered.

Now the geometry, which is where the proof lives. Put the two switch positions on a graph: first switch along the horizontal axis (0 = down, 1 = up), second along the vertical. Four corner points result: (0,0) and (1,1) should output OFF, while (0,1) and (1,0) should output ON.

A single-layer perceptron, when you unpack its weighted-sum-and-threshold arithmetic, can only ever do one geometric thing: **draw one straight line** through that picture and answer ON for every point on one side, OFF for the other. For the function AND (on only when both switches are up), a single line works: isolate the corner (1,1). For OR, likewise: isolate the corner (0,0). Now try XOR: you must put (0,1) and (1,0), two **diagonally opposite** corners, on one side of a straight line, and the other two diagonal corners, (0,0) and (1,1), on the other side.

TRY IT YOURSELF

Do it on paper now; it takes ninety seconds and makes the theorem unforgettable. Draw the unit square, mark the two ON corners and the two OFF corners, and try to draw one straight line separating ON from OFF. Every attempt strands a corner on the wrong side. That impossibility, which you have just verified by hand, is the celebrated result: **a single-layer perceptron cannot learn XOR**, because no single straight line separates its cases. The convergence theorem's fine print, "if the examples can be separated", has just bitten.

6.1.2 The tragedy in the fine print

Here is the part that turns a theorem into a tragedy. Add a **hidden middle layer** of neurons between input and output and the limitation evaporates: a two-layer network solves XOR trivially, essentially by letting intermediate neurons compute simpler pieces (like OR and NOT-AND) that the output neuron combines. Minsky and Papert knew this and said so. But they also observed, correctly for 1969, that **nobody knew how to train** multi-layer networks: Rosenblatt's learning rule tells you how to blame the output neuron's dials for an error, but offers no way to apportion blame backward through hidden layers. That missing bookkeeping, the **credit assignment problem**, would not be solved in a practical, popularised form until **backpropagation** in 1986, seventeen years later, and is precisely what you will implement in Week 3.

So the field faced a correct criticism of the simplest model, an acknowledged escape route, and no map to the escape route. What happened next was sociology, not mathematics: funders and department heads compressed the nuance into "neural networks are a dead end", money and students drained away for nearly two decades, and Rosenblatt, the approach's champion, died in a sailing accident in 1971 without seeing any vindication.

EVERYDAY ANALOGY

It is as if someone published a rigorous proof that a **bicycle** cannot climb stairs, mentioned in passing that a vehicle with more wheels and a motor probably could, and the world concluded that **wheels** were refuted. True about that bicycle; silent about wheels. When you meet criticisms of today's systems (Chapter 13 covers several serious ones), the XOR episode supplies the discipline: always ask **which version** of the idea a criticism actually reaches, and whether an acknowledged escape route exists.

6.2 The economic trigger: promises come due

Meanwhile the symbolic mainstream was defaulting on its own loans. Machine translation, generously funded through the Cold War for obvious reasons, was reviewed in 1966 by the ALPAC committee, which found it slower, worse, and more expensive than human translators; funding was effectively ended. (The era's cautionary legend, likely apocryphal but too instructive to omit: "the spirit is willing but the flesh is weak", machine-translated to Russian and back, returning as "the vodka is good but the meat is rotten." The story survives because it names the real problem: translation requires **meaning**, not word substitution.)

In Britain, the applied mathematician Sir James Lighthill was commissioned to review the field, and his 1973 report was devastating: in every area, he wrote, AI's discoveries had fallen far short of what was promised, and its methods collapsed outside toy problems. British AI funding was gutted to a couple of sites. Lighthill and McCarthy debated on BBC television, a rare spectacle of a scientific field defending its existence in prime time, and the search term "Lighthill debate" will find it for you; it is a time capsule. In the United States, ARPA's successor DARPA moved from funding people to funding deliverables, with comparable effect.

6.2.1 The real technical villain: combinatorial explosion

Beneath the politics lay one honest technical cause, and it is a phrase this module will use for twelve weeks. Golden-years methods worked by **search**: explore possibilities until something works. In a microworld, possibilities are few. In the real world, they **multiply**: every added object, word, or move multiplies the branches, so the search space grows **exponentially**, and no computer, then or now or ever, outruns exponential growth by brute force alone.

EVERYDAY ANALOGY

Password cracking makes the growth visceral. A four-digit PIN has ten thousand possibilities: trivial. Each added digit multiplies by ten. A sixteen-character password over a rich alphabet has more possibilities than there are atoms in a planet: hopeless, **and the difference was only twelve characters**. Exponential growth is not "harder"; past a small threshold it is a wall. Golden-years AI kept meeting that wall a few steps outside every microworld.

KEY IDEA

"It does not scale" is the most important critical phrase in this module. The first winter taught its original meaning: methods that shine on small cases can be **structurally** incapable of surviving growth, not merely slow. Watch the phrase recur: expert systems fail to scale in knowledge maintenance (Chapter 7), shallow models fail to scale in capability (Chapter 9), and, in the module thesis, brute-force scaling of language models is itself now hitting walls of cost, energy, and data (Chapter 13). One era's winning move is the next era's scaling failure.

❄ The First AI Winter, roughly 1974 to 1980: anatomy

Trigger, intellectual: the Minsky-Papert critique freezes neural network research. **Trigger, economic:** ALPAC (1966) and Lighthill (1973) document the gap between promises and results; DARPA and UK funding collapse. **Underlying cause:** combinatorial explosion; microworld success did not extrapolate. **Casualties:** machine translation, neural networks, general problem-solving programmes, and the field's credibility. **What survived:** the ideas. Theorem proving, planning, and knowledge representation continued quietly; the perceptron's learning loop waited for backpropagation. **Lesson for the pattern:** winters punish **promises**, not science. Nothing in the underlying mathematics was refuted; the timelines were.

QUIZ WATCH · WEEK 6

The XOR question is close to guaranteed in some form. Be able to: explain XOR via the stairwell light; explain **geometrically** why one layer fails (one straight line, diagonal corners); state the escape route (hidden layer) and the reason it was blocked in 1969 (no training method for hidden layers, solved by backpropagation, 1986); and, for full marks, say why the episode was an overreaction, using the bicycle analogy.

7 The Expert Systems Boom: 1980 to 1987

COVERS	AI's first commercial era: how expert systems worked, their real victories, and the three diseases that killed them
TIME	About 25 minutes
IN BRIEF	The field revived by narrowing its ambition: capture one expert's knowledge in one domain as hand-written rules, and sell it. It genuinely worked, made real money, and then failed for three reasons every engineer should be able to recite.

Fields recover from winters the way towns recover from busts: with a smaller, more sellable promise. The promise of the 1980s was this: forget general intelligence; instead, capture what one **human expert** knows about one **narrow domain**, encode it as explicit rules, and sell the result to industry. The product was called an **expert system**, and for about seven years it was the most successful idea in the field's history.

7.1 How an expert system works

The architecture has two parts, and the separation was the era's proudest engineering idea. The **knowledge base** holds domain know-how as hundreds or thousands of IF-THEN rules, written by "knowledge engineers" who interview human experts:

> IF the infection is bacterial AND the patient is allergic to penicillin THEN recommend erythromycin.
> IF the stain is gram-positive AND the morphology is coccus AND growth is in chains THEN the organism is likely streptococcus.

The **inference engine** is the general machinery that chains rules together: given the facts of a case, it fires whichever rules match, adds their conclusions to the facts, and repeats until it reaches a recommendation. Because the engine is generic, the same software shell could, in principle, become a medical adviser or a geologist by swapping knowledge bases; an industry of "expert system shells" sold exactly that promise.

Notice the family resemblance: this is **symbolic AI** (Chapter 4) in its purest commercial form, and it inherits the family's signature virtue. An expert system can **show its work**: ask MYCIN why it suspects streptococcus and it lists the rules it fired. Transparency of exactly this kind returns as an ethical requirement in Week 2, where the modern difficulty is that today's best systems, deep networks, conspicuously lack it.

7.2 The victories were real

It is important to internalise that this boom was not smoke. **DENDRAL** (Stanford, begun 1965) inferred molecular structures from mass-spectrometry data at expert level, arguably the first program to match human specialists at a genuine scientific task. **MYCIN** (Stanford, 1970s) diagnosed blood infections and recommended antibiotics from about six hundred rules, and in a formal evaluation its recommendations were rated acceptable at a rate that matched or exceeded the human specialists it was compared against. It was never deployed clinically, for reasons worth pausing on: liability (who answers for a machine's fatal mistake?), integration into clinical workflow, and physician trust. Note the shape of that sentence: the **technical** system succeeded and the **sociotechnical** system around it, responsibility, workflow, trust, did not exist. That is a Week 2 lecture in one historical anecdote, and a pattern you will meet in every AI deployment story of the 2020s.

The money came with **XCON** (also called R1), built with Carnegie Mellon for Digital Equipment Corporation from 1978. DEC sold minicomputers in bewildering configurations, and human order-checking was slow and error-prone. XCON, growing to thousands of rules, configured orders automatically and was credited with saving

DEC on the order of forty million dollars a year by 1986. Real product, real balance sheet, real annual savings: AI's first unambiguous commercial win.

The boom followed. By the mid 1980s a large fraction of major corporations had AI groups; venture capital arrived; Japan announced its **Fifth Generation Computer Systems** project, a decade-scale national programme to leapfrog the West via massively parallel logic machines, and the US and Britain launched counter-programmes in response (fear of a rival's national AI project producing sudden domestic funding is a dynamic you may recognise from the 2020s). A hardware industry grew up selling **Lisp machines**: expensive workstations built to run the field's native language (Chapter 4) fast. AI industry revenue climbed into the billions.

7.3 The three diseases

By 1987 the boom was dead. Not from one blow, but from three chronic conditions that every rules-based approach carries, and that scale made fatal. These three are among the most examinable items in this booklet, because each teaches a permanent engineering lesson.

7.3.1 Disease one: brittleness

An expert system knows **only** its rules. Step one inch outside what its authors foresaw and it does not degrade gracefully; it fails absurdly, and, worse, **confidently**, because nothing in the architecture represents the boundary of its own competence. The medical system asked about a rusty car will happily attempt a diagnosis. Systems of the era famously lacked what critics called common sense: MYCIN could out-reason physicians about blood cultures while not knowing, in any sense, what blood **is**.

> **EVERYDAY ANALOGY**
>
> A phrasebook tourist. With the book in hand, they order dinner flawlessly and ask directions like a local. The moment a waiter improvises ("we're out of the fish, but the chef can do it with hake?"), the performance collapses, because there was never any understanding underneath, only lookup. Rules are a phrasebook: exhaustive for the anticipated, empty for everything else. And reality, as the first winter already taught, is mostly the unanticipated.

7.3.2 Disease two: the knowledge bottleneck

The business model assumed experts could **tell** knowledge engineers what they know. They largely cannot. Ask a chess grandmaster why a move is right: "it felt strong." Ask an experienced clinician how they knew a patient was deteriorating before the numbers moved: "she looked wrong." Decades of expertise compile into fast, unconscious pattern recognition, what the philosopher Michael Polanyi called **tacit knowledge**, summarised in his line that **we know more than we can tell**. Interviewing cannot extract what introspection cannot reach, so knowledge bases plateaued at what experts could articulate, which is the smaller half of expertise.

> **EVERYDAY ANALOGY**
>
> Ask your grandmother how much salt goes in the stew. "Until it tastes right." She is not withholding the rule; **there is no rule to withhold**, only a judgement trained by ten thousand tastings. Now the punchline that organises this whole module: machine learning's foundational wager is that tacit knowledge can be absorbed **from examples** instead of extracted **by interview**. Show a network ten thousand labelled stews, or tumours, or chess positions, and it learns the judgement directly, exactly as grandmother did. The knowledge bottleneck is the specific disease that learning from data cures, which is why data-driven AI eventually displaced rules nearly everywhere.

7.3.3 Disease three: maintenance economics

Thousands of hand-written rules interact. Add rule 3,001 and it can silently contradict rule 214 in cases nobody imagined; domain knowledge also **changes**, so the base rots without constant expert-hours. XCON at its height needed a large permanent staff simply to keep it true. Costs grew **with the square of ambition**, roughly, since every new rule must coexist with all existing ones, while the value grew linearly. That arithmetic has an ending, and by the late 1980s corporate accountants had found it.

KEY IDEA

The three diseases, brittleness, the knowledge bottleneck, and maintenance economics, are one lesson wearing three coats: **hand-written rules do not scale**, in robustness, in acquisition, or in upkeep. This is the second historical meaning of "does not scale" (Chapter 6 gave the first), and it is the specific failure that makes learning from data not merely fashionable but **structurally necessary**. When a stakeholder someday asks you "why can't we just write rules for this?", your answer is this section, compressed: rules break outside anticipation, experts cannot articulate their expertise, and rule bases rot faster than they can be repaired, **whenever the domain is open-ended**. (For small, closed, stable domains, rules remain exactly the right tool; tax tables need no neural network. The judgement is the skill.)

QUIZ WATCH · WEEK 6

Near-certain territory: name and explain the three reasons expert systems collapsed, with an analogy each (phrasebook tourist, grandmother's salt, and rot-versus-repair). The follow-up worth preparing: "which disease does machine learning specifically cure, and how?" (The bottleneck: learning absorbs tacit knowledge from examples rather than interviews.)

8 The Second Winter and the Great Rebrand: 1987 to 1993

COVERS The collapse of the AI industry, the death of the Lisp machine, and the rebrand that renamed the field

TIME About 15 minutes

IN BRIEF The second winter was economic where the first was intellectual: a hardware market evaporated, corporate AI groups closed, and "artificial intelligence" became a phrase researchers avoided. The field survived by renaming itself.

The second winter arrived faster than the first and hit harder, because this time there was an **industry** to destroy, with payrolls and shareholders.

8.1 The collapse

The proximate blow was hardware. The Lisp machine industry existed because ordinary computers of the late 1970s ran the field's software poorly; specialised workstations closed that gap at a premium price. But general-purpose computing was compounding relentlessly (Moore's law, in effect), and by 1987 desktop machines from Sun and Apple ran Lisp software as fast as the specialist hardware at a fraction of the cost. A half-billion-dollar market disappeared within a couple of years; its flagship firms went bankrupt or fled the business.

The software side followed as the three diseases presented their invoices. Expert systems that had dazzled in pilots proved brittle and ruinous to maintain at scale; corporate AI groups, launched in the mid-80s exuberance, quietly closed by the turn of the decade. Japan's Fifth Generation project wound down having met almost none of its headline goals (honourably: its concrete predictions made its shortfall measurable, and its investment in parallel computing bore fruit elsewhere). DARPA cut strategically. The pattern of Chapter 6 replayed at industrial scale: promises due, results short, capital gone.

8.2 The rebrand, and why names matter

The distinctive scar of the second winter was linguistic. "Artificial intelligence" became a mark against funding proposals and business plans, a phrase associated with two crashes in fifteen years. Researchers responded rationally: they kept doing the work and **changed what they called it**. Grant applications and job titles bloomed with **machine learning**, **pattern recognition**, **informatics**, **knowledge-based systems**, **data mining**, and, with a straight face, **statistics**.

The rebrand was camouflage, but, and this is the important part, it was also **honest signalling of a real intellectual shift**. The name "machine learning" foregrounded exactly the thing the expert systems era had lacked and the perceptron had promised: systems that acquire behaviour from data. The field that emerged from winter under the new name really was different, as the next chapter shows: probabilistic, statistical, evaluation-driven, and allergic to grand claims. When "artificial intelligence" returned as a respectable label in the 2010s boom, it was as a marketing umbrella over techniques developed under the modest names; this module's own title, with "Machine Learning" first, is a fossil of that history.

Surviving the winter: an unfashionable trio

Through both winters, a small group kept working on neural networks when doing so was career poison, and their persistence is the hinge of this entire history. **Geoffrey Hinton**, in Britain then Canada, co-authored the 1986 paper that popularised **backpropagation**, the missing training method for multi-layer networks, the escape route from XOR (Chapter 6), arriving seventeen years after the door was pointed out. **Yann**

LeCun, in France then at Bell Labs, applied it in convolutional networks that by the late 1990s were reading a meaningful share of cheques deposited in the United States: deep learning was in industrial production a decade before it was famous, a fact worth deploying whenever someone calls the field pure hype. **Yoshua Bengio** in Montreal pushed neural approaches to language and probability. Canadian public funding (notably the CIFAR institute) kept the flame lit when US agencies would not; the geography of the modern AI industry, with its strong Toronto and Montreal poles, is partly downstream of who was willing to fund unfashionable ideas in 1990. The trio shared the 2018 Turing Award, and Hinton later shared a Nobel Prize. Moral for the pattern: **a dormant field and a dead field look identical from outside; winters punish popularity, not truth.**

❄️ The Second AI Winter, roughly 1987 to 1993: anatomy

Trigger, economic: Lisp machine market collapses under commodity hardware; expert systems' maintenance costs come due; corporate and government funding retreats. **Trigger, intellectual:** the three diseases of rules-based AI become undeniable at scale. **Casualties:** the first AI industry, the Fifth Generation project's ambitions, and the very name "artificial intelligence". **What survived:** everything important, renamed. Backpropagation (1986) arrived **during** the collapse; statistical methods matured in the winter's shadow. **Lesson for the pattern:** this winter punished a **business model** (hand-written knowledge at scale) and a **hardware bet** (specialist machines against commodity curves). Both lessons are live: whenever you evaluate an AI venture built on bespoke hardware or manually curated knowledge, you are re-running 1987's due diligence.

QUIZ WATCH · WEEK 6

Be able to contrast the two winters in one tight paragraph: the first was primarily **intellectual** (a critique plus overpromising, punished by research funders), the second primarily **economic** (an industry's collapse, punished by markets), and both were caused by the same underlying gap between promises and scalable results. Add the rebrand and its double nature (camouflage **and** honest signal) for full marks.

9 The Quiet Revolution: The Statistical Turn, 1990s to 2011

COVERS How the field rebuilt on probability and data: everyday ML you already used, Deep Blue, the AI effect, and the keepers of the flame

TIME About 20 minutes

IN BRIEF While headlines said AI was dead, it colonised daily life under other names: spam filters, credit scoring, search ranking, recommendations. The paradigm quietly flipped from writing rules to learning patterns, and the stage assembled itself for 2012.

The most productive period in the field's history is the one the public never noticed. Between the early 1990s and 2011, "dead" AI moved into your mailbox, your bank, and your search results, because the second winter's rebrand came with a genuine change of method.

9.1 The paradigm flips

The new discipline of machine learning inverted the expert systems recipe precisely. Instead of interviewing experts and writing rules, you gather **labelled examples** and let an algorithm find the patterns. Nobody wrote "IF the email contains FREE MONEY THEN spam"; the filter learned, from millions of user-flagged messages, which features predict junk, and, crucially, **kept learning** as spammers adapted, a moving target no hand-written rule base could chase (recall disease three). The same inversion powered credit scoring, fraud detection (your card's uncanny sense of which purchases are not you), product recommendations, and the ranking behind web search. By 2005, an ordinary person interacted with learned models dozens of times a day while believing AI had failed. Keep that gap in mind; public perception of this field routinely runs a decade behind deployment, in both directions.

The intellectual style changed with the method. The era's native mathematics was probability and statistics rather than logic: reasoning under uncertainty (Judea Pearl's Bayesian networks made this rigorous, earning a Turing Award), margins and generalisation theory (support vector machines, the era's flagship algorithm, which you will meet as a working tool in Week 3 alongside decision trees and regression). And the culture changed, in a way traceable straight back to Turing's test: shared datasets and **benchmarks** replaced demonstrations. Claims now required numbers on a common test set that a rival could check. The era's grand claims allergy was winter-taught humility, and its benchmark habit is the specific cultural machinery that will make 2012 possible.

9.2 Deep Blue, 1997: the old paradigm's finest hour

The decade's famous AI moment was, ironically, not machine learning at all. In May 1997, IBM's **Deep Blue** defeated the reigning world chess champion, Garry Kasparov, three and a half to two and a half, in a match watched worldwide. Simon and Newell's 1957 ten-year prediction (Chapter 4) had come true, forty years late, which is the field's forecasting record in a single sentence.

Understand **how** it won, because the how is the point. Deep Blue was custom hardware evaluating on the order of two hundred million chess positions per second, steered by hand-tuned evaluation knowledge partly derived from grandmaster consultation. It was brute-force search plus expert rules: the golden years' method and the eighties' method, fused and perfected, aimed at the one arena where they could still win, a **closed** world with fixed rules and perfect information, exactly the swimming pool (Chapter 5) with the rope removed. It was magnificent, and it was a dead end: nothing in Deep Blue could recognise a face, parse a sentence, or transfer one iota of its chess mastery to any second task. The future belonged to the unglamorous statistical work happening everywhere else, and the contrast between Deep Blue and AlphaGo (Chapter 11) will make the paradigm shift measurable.

Kasparov, the ghost in the machine, and a lesson about opacity

During game one, Deep Blue played a strange move that Kasparov could not attribute to calculation; brooding on it, he began to suspect either superhuman depth or human interference, and many analysts believe the psychological shadow of that move contributed to his uncharacteristic collapse in game two and the match. Years later, members of the Deep Blue team explained the move as most likely the output of a **bug**, a fallback behaviour when the machine could not resolve its search. Savour every layer: the era's most famous demonstration of machine intelligence turned partly on a human **projecting deep intent onto a malfunction**, the ELIZA effect (Chapter 5) operating at world-championship level. When Week 2 discusses why opaque systems require interpretability tools, remember that even Kasparov, facing a mere chess program, could not distinguish brilliance from breakage, and paid for the confusion.

9.3 The AI effect: moving goalposts, in both directions

The public reaction to Deep Blue introduced a pattern with its own name. Within days, commentary settled on a dismissal: it was "just brute-force search", not **real** intelligence. Chess, which for a century had been literature's shorthand for genius, was reclassified as mere calculation, **at the moment a machine mastered it**.

This is the **AI effect**: whenever AI achieves a milestone, the milestone is redefined as not-really-intelligence. It has since consumed translation, image recognition, Go, and conversational fluency in turn. Two things are true about it at once, and holding both is the calibration this module keeps asking of you. The dismissals contain real signal: Deep Blue's method genuinely was narrow, and noticing that is analysis, not cope. And the pattern, applied relentlessly, becomes unfalsifiable: if every achieved goal is retroactively downgraded, "real AI" means "whatever machines cannot yet do", a definition that can never be met.

TRY IT YOURSELF

Calibration exercise, five minutes, excellent forum material: write down, privately and **now**, a capability such that, if a machine demonstrated it, you would accept the machine was genuinely intelligent. Date the note. Check it at Week 12, and notice whether you moved your own goalposts during the semester. Most people do, in one direction or the other; the point of the exercise is to catch yourself doing it.

KEY IDEA

The statistical turn is the pivot of this whole history: the moment the field's centre of gravity moved from **rules written by hand** to **patterns learned from data**, from logic to probability, and from demonstrations to benchmarks. Every ingredient of the modern era except scale is in place by 2005. What is missing is a way for learned models to be **deep**, and the three supplies, data, compute, architecture, that the next chapter assembles.

QUIZ WATCH · WEEK 6

Be ready to: name three everyday 1990s-2000s products that were "secretly ML" and say what replaced rules in each (learned patterns from labelled data); explain why Deep Blue is best classed with the **old** paradigm despite its date; and define the AI effect with one example and one sentence on why it is a calibration trap in both directions.

10 The Big Bang: Three Ingredients and AlexNet, 2012

COVERS The single most important chapter: why deep learning ignited in 2012 and not before, told as a supply story of data, compute, and architecture

TIME About 25 minutes

IN BRIEF The 2012 breakthrough was not a new idea. It was the moment three separately maturing supplies, big labelled data, cheap parallel compute, and trainable deep architectures, finally coexisted. Master this chapter and the whole history snaps into focus.

If this booklet has a centre, it is here. Everything before this chapter explains why the ingredients were missing; everything after is what happened once they combined. The claim to be tested and internalised is deliberately provocative: **the deep learning revolution involved almost no new ideas.** It was a revolution of supply.

10.1 Ingredient one: data, and the unfashionable bet behind it

A learning system's ceiling is its experience. The perceptron era owned hundreds of examples; the statistical era, thousands to low millions of narrow records. Deep networks, with their millions of adjustable dials, need experience on another scale entirely, and until the late 2000s it simply did not exist in usable form.

The internet created the raw material; turning it into **labelled** training data took a contrarian act of infrastructure-building. Around 2006, Fei-Fei Li, then a young professor, judged that the field was obsessing over algorithms while starving of data, and set out to label a meaningful slice of the visual world. The project, **ImageNet**, eventually organised over fourteen million photographs into tens of thousands of categories, built by distributing the labelling work to tens of thousands of crowdworkers through Amazon's Mechanical Turk platform (whose name, recall from Chapter 2, winks at an eighteenth-century fake automaton powered by hidden humans; the joke is that this time the hidden humans were the point). Colleagues advised her against it: datasets were service work, not science, and no model of the day could exploit such a corpus anyway. From 2010 ImageNet ran an annual public recognition challenge, the benchmark culture of Chapter 9 operating at full strength, and thereby built the arena in which the revolution would be scored.

10.2 Ingredient two: compute, by way of the games industry

Training a network is arithmetic: enormous numbers of small multiplications and additions, most of them independent of one another and therefore performable **simultaneously**. Ordinary processors (CPUs) are a few brilliant workers; the arithmetic of learning wants ten thousand adequate ones.

That machine existed, built for an entirely different customer. The **graphics processing unit** had been developed across two decades to render video game imagery, a task which is **also** millions of small independent arithmetic operations done simultaneously (one per pixel, roughly). Decades of gamers' spending had driven a ferocious price-performance curve, and when NVIDIA released CUDA in 2007, opening its chips to general-purpose programming, researchers discovered that the ideal neural network trainer had been sitting in teenagers' bedrooms all along. Speedups of one to two orders of magnitude arrived essentially overnight, at consumer prices.

EVERYDAY ANALOGY

The GPU story is the **adjacent industry subsidy**: transformative infrastructure often arrives pre-built, paid for by a neighbouring market that wanted it for other reasons, like railways built for coal that end up carrying passengers. Nobody funding "AI" could have justified two decades of chip development; the games industry did it for fun. Two professional habits follow. When evaluating why a technology is suddenly feasible, ask **whose spending built the road**. And when forecasting, scan adjacent industries for roads being built right now; Week 9′s edge-AI story, riding on the smartphone industry's chips, is the same pattern in progress.

10.3 Ingredient three: architecture, mostly waiting in a drawer

The third supply is the one with the longest pedigree, which is exactly the point. **Backpropagation**, the method for training multi-layer networks (the escape route from XOR), was popularised in 1986. The **convolutional network**, an architecture wiring in the assumption that images have local, position-independent structure, was matured by LeCun around 1989-1998 and, as Chapter 8 noted, read cheques industrially in the 1990s. (Architectural assumptions of this kind, structure as a claim about the world, are precisely the subject of Week 4, and Week 5 extends them to physical law.) What the 2000s added were practical unblockers, modest-looking tricks that made **deep** training stable where it had previously stalled: a better activation function (ReLU), a simple anti-overfitting method (dropout), and Hinton's 2006 pre-training results, which mattered as much for morale as mathematics: they re-legitimised depth, and rebranded the unfashionable field as **deep learning**.

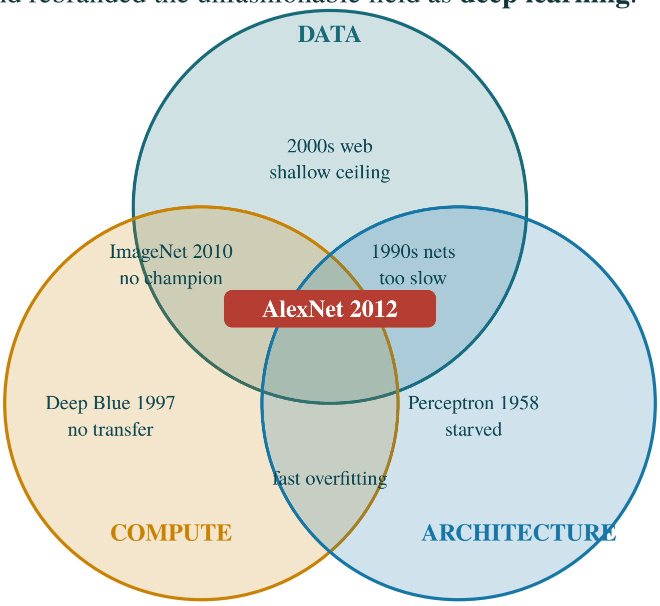

Figure 10.1 · Every region has a name and a date attached; only the centre has a field in it. Partial combinations are not near misses, they are eras.

10.4 The thesis, as an experiment you can run

The three-ingredient claim is historical, so it is easy to state and hard to test. The Week 1 lab turns it into a measurement. `three_ingredients.py` sets data (number of training examples), architecture (hidden units) and compute (optimiser iterations) to LOW or HIGH independently, and reports held-out accuracy for all eight combinations, on the digits dataset that ships with scikit-learn. No GPU, no download, about a minute:

```
data  arch  comp  n_train   hidden  iters  train   TEST  era
 low   low   low      120     (2,)      8  0.189  0.190  nothing at all
HIGH   low   low     1257     (2,)      8  0.405  0.402  data alone: the 2000s web
```

```
  low   low  HIGH      120      (2,)    600  0.719  0.451  compute alone: Deep Blue 1997
 HIGH   low  HIGH     1257      (2,)    600  0.708  0.672  data+compute: ImageNet 2010
  low  HIGH   low      120 (128, 64)      8  1.000  0.862  architecture alone: perceptron
  low  HIGH  HIGH      120 (128, 64)    600  1.000  0.881  arch+compute: fast overfitting
 HIGH  HIGH   low     1257 (128, 64)      8  0.999  0.962  data+arch: 1990s, too slow
 HIGH  HIGH  HIGH     1257 (128, 64)    600  1.000  0.970  ALL THREE: AlexNet, 2012
```

Read the era column against the accuracy column. Each partial combination stalls roughly where history says it stalled, and no single ingredient carries the result. Note row five in particular: training accuracy 1.000 against test accuracy 0.862. A large model with a small dataset does not fail to learn, it **memorises**, which is the 1990s in one line and Week 3′s overfitting lecture arriving early.

> **TRY IT YOURSELF**
>
> Find a setting in which **more compute makes test accuracy worse**, and explain it in one sentence using a term from Week 3. Then argue with the `SUPPLY` dates in `timeline_data.py`: they claim architecture arrived in 1986, compute in 2007 and data in 2009, which puts all three in place by 2009 and leaves a three-year gap before anyone assembled them. Defend your own dates in three sentences. There is no correct answer here; there are defensible ones.

10.5 September 2012: the ingredients meet

At the 2012 ImageNet challenge, a network called **AlexNet**, built by Alex Krizhevsky with Ilya Sutskever under Hinton at Toronto, a lab funded through the winter by Canadian money (Chapter 8′s story pays off here), combined the three supplies for the first time at full strength: fourteen million labelled images, **two consumer gaming GPUs** in a desktop, and an eight-layer convolutional network using the new unblockers.

The scoreboard did the arguing. The previous year's winning error rate was around 26 percent; contests of this maturity normally improve by fractions of a point. AlexNet scored roughly 15 percent, a ten-point demolition of the field, with the runner-up nowhere. Within one cycle, every serious entrant was a deep network; within about three years, contest error fell below typical human performance on the task, and the method had spread from vision to speech and beyond. Almost nothing in AlexNet was conceptually new; nearly every component existed by 1998. What was new was that the world could finally **afford to run the old ideas at scale**, and the benchmark culture ensured the result was public, quantified, and impossible to dismiss.

> **KEY IDEA**
>
> **The three-ingredient thesis**, the single most important idea in Week 1: deep learning requires big labelled **data**, cheap parallel **compute**, and trainable deep **architectures**, simultaneously. Every partial combination had already been tried and had stalled, and you can name the eras: architecture without data or compute is the perceptron era (right idea, starved); compute without learning is Deep Blue (power without transfer); data feeding shallow models is the 2000s web (value with a low ceiling); data plus compute awaiting architecture is ImageNet 2010-2011 (an arena with no champion). The breakthrough was a **coincidence of supply**, and its ideas were mostly decades old. Corollary for your future forecasting: the next breakthrough may likewise be an old idea whose missing ingredient just arrived; the interesting question about any stalled approach is **which supply it is waiting for**.

> **TRY IT YOURSELF**
>
> The Interactive Companion's three-ingredient experiment is this chapter as a toy: toggle Data, Compute, and Architecture and check that you can predict, **before clicking**, which historical era each of the eight combinations lands in, and why. If you can narrate all eight from memory, this chapter is revision-complete.

QUIZ WATCH · WEEK 6

Expect the three ingredients in some form, very possibly as the essay-style item. Be able to: name all three with a plain-English gloss each; attach a stalled era to at least two partial combinations; explain the GPU supply via the games industry; explain ImageNet's role including the benchmark culture; and defend, in three sentences, the claim "2012 was a coincidence of supply, not a new idea".

11 From Vision to Everything: 2012 to 2017

COVERS	The method spreads: speech, translation, generation, and AlphaGo's learned intuition
TIME	About 15 minutes
IN BRIEF	Five years took deep learning from a vision contest to superhuman Go, and from recognising content to **generating** it. AlphaGo against Deep Blue is the cleanest before-and-after picture of the paradigm shift.

Revolutions are measured by how fast the method escapes its home domain. Deep learning's escape was immediate and total: speech recognition error rates collapsed (the reason phone dictation abruptly became usable mid-decade), machine translation was rebuilt end-to-end around learned sequence models with quality jumps users noticed overnight, and the major technology companies pivoted hard, hiring the winter's survivors into industrial labs (Hinton to Google, LeCun to head a new lab at Meta, then Facebook) and open-sourcing frameworks, culminating in PyTorch (2016), the tool of your Week 3 labs, in which the loop you will write, forward pass, loss, backward pass, update, is the darts loop of Chapter 5 industrialised.

Generation arrived alongside recognition. **Generative adversarial networks** (Goodfellow, 2014) trained a forger network against a detective network, each improving the other, and photorealistic synthesis of nonexistent faces followed within a few years; **diffusion models**, which later displaced GANs and power the image generators you have used, take a different route (learning to reverse gradual noising) with a deep connection to thermodynamics that Week 5 treats as a serious scientific idea rather than a curiosity. Generation also delivered ethics its modern urgency: recognising a face and **fabricating** one are different moral objects, and Week 2′s fairness and provenance material stands on this ground.

11.1 AlphaGo: the paradigm shift, scored

The era's defining public event replayed 1997 with the paradigm inverted. The board game Go had been the standing counterexample to Deep Blue's method: with roughly 250 reasonable moves per turn to chess's 35, and games twice as long, its search space (about ten to the power 170 positions, dwarfing the atoms in the observable universe, which number around ten to the power 80) put brute force permanently out of reach; combinatorial explosion (Chapter 6) had won, and professionals spoke of good moves in the vocabulary of **aesthetics and intuition**, precisely the tacit knowledge (Chapter 7) that cannot be written as rules.

DeepMind's **AlphaGo** met tacit knowledge with the cure from Chapter 7: it **learned** judgement from examples, thirty million positions from strong human games, then improved beyond its teachers through **self-play**, generating fresh experience by the million against copies of itself, using learned evaluation to steer a comparatively modest search. In March 2016 it defeated Lee Sedol, among the greatest players of the era, four games to one, before an audience estimated in the hundreds of millions, in a region where Go carries deep cultural weight.

The match produced the period's most quoted single move. In game two, AlphaGo's **move 37** violated centuries of professional principle, so far outside expectation that commentators initially suspected a malfunction (note the rhyme with Kasparov's ghost-move, inverted: there, a human read genius into a bug; here, humans read a bug into genius). It proved pivotal, and Lee's own **move 78** in game four, a "divine move" that beat the machine once, belongs in the same story. Machines that learn can **originate**, Lovelace's objection (Chapter 2) meeting its strongest empirical answer, and can be beaten by human creativity on the same board, in the same week.

KEY IDEA

Deep Blue versus AlphaGo is the paradigm shift in one comparison, and a near-certain exam contrast. Deep Blue: hand-tuned knowledge plus brute-force search; conquered a searchable closed world; transferred nothing. AlphaGo: **learned** evaluation (from human examples, then self-play) steering modest search; conquered a world where brute force is impossible **because** judgement there is tacit; and its successor AlphaZero, learning three different games from bare rules with one method, demonstrated the transfer that Deep Blue structurally lacked. Same genre of arena, opposite theory of intelligence, nineteen years apart.

QUIZ WATCH · WEEK 6

Prepare the Deep Blue / AlphaGo contrast as a table you can reproduce: method, why suited to its game, role of human knowledge, transferability. Attach move 37 to the Lovelace objection in one sentence.

12 The Language Era: Transformers, Scaling, and ChatGPT, 2017 to 2023

COVERS Attention in plain language, the GPT lineage, scaling laws, emergence, RLHF, and the night AI met the public

TIME About 25 minutes

IN BRIEF One 2017 architecture, the transformer, made language learnable at scale; scaling laws made progress **purchasable**; and a 2022 interface decision made it public. This chapter carries you to the doorstep of the module thesis.

Vision fell first because images were the easy case: fixed-size grids with local structure, exactly what convolutions assume (Week 4′s theme, architecture as assumption). Language resisted longer because its structure is **relational and long-range**: the meaning of a word can depend on another word forty sentences back.

12.1 The transformer, 2017

Pre-2017 language models (recurrent networks, Week 4) read text the way you would read through a keyhole: one word at a time, carrying a compressed running memory, which blurred over distance and, worse for the coming era, forced **sequential** processing that wasted the GPU's ten thousand workers (Chapter 10). The 2017 Google paper **Attention Is All You Need**, title now legendary for underselling, proposed the **transformer**, which discards the keyhole entirely.

Its core mechanism, **attention**, lets every word, in effect, look at every other word in the passage simultaneously and learn **how much each matters** for interpreting it. In "the trophy would not fit in the suitcase because it was too big", resolving "it" requires weighing "trophy" against "suitcase" using the meaning of "big"; attention is the machinery that learns such weightings from data rather than rules (the pronoun rules that expert systems era linguists attempted are a Chapter 7 story with the usual ending).

EVERYDAY ANALOGY

Reading a mystery novel well means holding the stranger from chapter 2 in mind when the twist lands in chapter 20. The keyhole reader relies on an increasingly smudged memory of chapter 2; the attention reader keeps the whole book open on the desk and consults any page instantly. And because "consult every pair of pages" is massively parallel arithmetic, the transformer is also the architecture the GPU had been waiting for: the mystery-novel trick and the ten-thousand-workers trick are the **same** trick. Note, for the module thesis, the cost hiding in the analogy: every page consulting every page grows with the **square** of the book's length. That quadratic bill is precisely the bottleneck Week 8′s scale-free architectures (Mamba and relatives) exist to dodge.

Every headline model since, the GPT series, Claude, Gemini, Llama, DeepSeek, is a transformer variant, trained on an objective almost embarrassing in its simplicity: **predict the next word**, over internet-scale text. The wager is that predicting text well enough forces a model to absorb, implicitly, much of what the text is about: grammar, facts, styles, and some degree of the reasoning that produced it. How far "some degree" extends is the field's central live argument, and this module's Week 7.

12.2 Scaling laws: progress becomes purchasable

Around 2020, OpenAI researchers quantified something practitioners had sensed: language model performance improves **smoothly and predictably** as you jointly grow three quantities, model size, dataset size, and training compute, following remarkably clean mathematical curves (power laws) that held across many orders of magnitude. Read the finding's economic meaning slowly, because it explains the next five years of industry behaviour: capability had become **forecastable and therefore purchasable**. A lab could project the ability of a model costing ten times more **before spending the money**, converting research risk into capital expenditure, and the arms race that followed, training runs climbing from millions toward hundreds of millions of dollars, was the rational response of every player to the same graph. When Chapter 13 presents the walls, they are the walls of **this** recipe.

GPT-3 (2020, 175 billion parameters) made the curves tangible and delivered the era's genuine surprise: **in-context learning**. Shown a few examples of a task inside its prompt, the model performs the task, with no retraining, despite never being explicitly taught to do so. Abilities of this kind, appearing at scale without being individually engineered, were labelled **emergent**, a term to handle with tongs: whether emergence is a deep fact about scale or partly an artefact of how abilities are measured is genuinely contested, and Week 7 gives the debate its due. The honest Week 1 statement: bigger models reliably did qualitatively more than smaller ones, ahead of detailed understanding of why, a situation, note, with no precedent anywhere earlier in this booklet, where capability **followed** understanding.

EVERYDAY ANALOGY

For emergence, the soup pot: enlarging the pot mostly gets you more soup, but past certain sizes, flavours appear that nobody added. For the epistemics, the honest cook's confession: we can **predict the flavours will come** (the scaling curves) without a full account of the chemistry (why **these** abilities at **this** scale). Engineering running ahead of science is historically common, steam engines preceded thermodynamics, but it is a new condition for AI, and it is one root of Week 2′s safety agenda.

12.3 ChatGPT, November 2022: the interface is the invention

The final step to the public was, technically, barely a step. The base model behind ChatGPT was a refinement of what had existed for two years. Two additions changed history. First, **RLHF**, reinforcement learning from human feedback: human raters compared candidate answers, a model of their preferences was trained, and the language model was tuned against it, teaching the raw text predictor to be **helpful, on-topic, and conversational**, closing the gap between "can complete any text" and "answers your question". (Note the lineage: the feedback-and-adjustment core of RLHF descends from reinforcement learning, cybernetics' goal-correction idea (Chapter 2), returning to centre stage seventy years on.) Second, a **chat box**: the frontier wrapped in the one interface every human already knew.

ChatGPT reached an estimated hundred million users in two months, the fastest consumer adoption then recorded, faster than any technology in this booklet by orders of magnitude. Governments, universities, newsrooms, and your relatives all acquired opinions about AI more or less simultaneously; the EU's AI Act, which Week 2 treats as engineering requirements rather than distant law, belongs to the world's reaction to this moment.

KEY IDEA

The invention people adopt is often the interface, not the engine. The engine predated ChatGPT by two years as a specialist curiosity; RLHF plus a chat box turned it into the fastest-spreading product in history. The lesson generalises across this booklet (the perceptron's press conference, ELIZA's couch, Deep Blue's theatre) and forward into your career: technical capability and adoption are separate achievements, and the second has its own engineering, which includes, as RLHF shows, **aligning** behaviour with human preference. It also has its own ethics: the ELIZA effect (Chapter 5) at hundred-million-user scale is no longer a curiosity but a design responsibility, which is Week 2′s opening argument.

QUIZ WATCH · WEEK 6

From this chapter: explain attention with the mystery-novel analogy **and** name the quadratic cost it hides (feeding Week 8); state what scaling laws showed and, in one sentence, their economic consequence; define emergence with the required epistemic caution; and explain the two additions that made ChatGPT from GPT-3 (RLHF, interface) with the general lesson.

13 The Walls and the Two Turns: 2023 to Now

COVERS Where the scaling recipe strains, the field's two responses, and the module thesis in full
TIME About 25 minutes
IN BRIEF The brute-force recipe is hitting four walls: cost, energy, data, and diminishing returns. The field's response is turning in two directions, grounding models in science and making them lean, and those two turns are the spine of the remaining eleven weeks.

Every previous era of this history ended when its winning method met a limit it could not out-spend. This chapter is about the current era arriving at that same moment, with you watching, which is why the module begins with history: the pattern gives you the instruments to read the present.

13.1 The four walls

Cost. Frontier training runs are estimated in the hundreds of millions of dollars each, before the still-larger cost of the hardware fleets and the engineers. The consequence is structural, not merely financial: a field that in 2012 could be advanced by a graduate student with two gaming cards (Chapter 10) now has a frontier open to a handful of companies and states, with everything that concentration implies for openness, scrutiny, and who sets the agenda. (Week 7 examines the open-versus-proprietary model landscape this has produced; Week 9′s lean techniques are, among other things, the counter-force that keeps meaningful work possible outside the giants, including in a university lab like yours.)

Energy. Training and, increasingly, **serving** models at global scale has made data-centre electricity demand a mainstream policy topic, discussed by grid operators and governments, not just conference workshops. Sustainability appears in Week 9 not as a virtue slide but as an engineering constraint with a budget attached.

Data. The scaling recipe consumed the accumulated text of the internet in about five years, and high-quality human writing is a **finite, slowly-renewing** resource; the frontier has visibly shifted toward synthetic data (models training on model output, with subtle failure modes), licensed private corpora, and non-text modalities. The supply that made the era (Chapter 10′s first ingredient) is the first to run short, a symmetry the exam may well ask you to notice.

Diminishing returns. The scaling curves (Chapter 12) are power laws: smooth, but **flattening**, so each equal jump in capability costs a multiple of the last. The curves have not broken; they have become expensive, which for a purchasable-progress industry is the same event in slow motion.

EVERYDAY ANALOGY

The four walls are the seaside town (Chapter 1) at the top of its third cycle, but with a difference worth respecting: this boom, unlike 1987′s, has enormous **paying** demand, hundreds of millions of daily users and real revenue, which is precisely what the expert systems economy lacked. A crash of promises remains possible; a disappearance of the technology does not. Calibration, as ever: the field can be simultaneously overhyped in its promises and permanent in its products, and both halves of that sentence have been true in every era of this booklet.

13.2 The first turn: grounding models in science

The first response attacks the data wall with knowledge. If a model must rediscover physics from pixels, it needs oceanic data; if the physics is **built in**, the data bill collapses. This is the oldest idea in the booklet wearing new mathematics: architecture as assumption (convolutions assume locality; Week 4) extended to assuming **physical**

law. Physics-informed neural networks bake governing equations into training itself, so the model is penalised for violating known science; equivariant and geometric networks wire in symmetries (a molecule's properties do not change when you rotate it, so why spend data teaching that?); neural operators learn solution maps for whole families of equations; and, in the direction of discovery rather than simulation, AlphaFold's protein-structure predictions, recognised with a Nobel Prize in Chemistry in 2024, demonstrated that learned systems can contribute at the level of scientific instruments. Week 5 is a full lecture and lab on these architectures (you will build a small physics-informed network and race it against a data-only baseline); Week 11 covers AI as an instrument of discovery. Notice, for your two-families bookkeeping (Chapter 4): building known equations into learning systems is symbolic knowledge re-entering connectionism through the front door, the pendulum becoming a synthesis.

13.3 The second turn: making models lean

The second response attacks the cost and energy walls directly: extract frontier-class capability from radically less. Its toolkit is Week 9 in miniature. **Distillation**: a small student model trained to imitate a large teacher, capturing much of the capability at a fraction of the size (tacit knowledge transferred machine-to-machine; grandmother's stew, one more time). **Quantization**: storing the dials (Chapter 5) at lower numeric precision, four bits or fewer instead of sixteen, shrinking memory and speeding inference for modest accuracy cost. **Parameter-efficient fine-tuning** (LoRA and relatives): adapting a huge model by training only a small attachment, bringing customisation within reach of a laptop. **Scale-free architectures** (Week 8): state-space models such as Mamba that dodge attention's quadratic bill (Chapter 12′s flagged cost) with linear-time, constant-memory designs. And a strategic novelty: **reasoning models** that improve answers by spending more computation **at answer time**, thinking longer rather than growing bigger, scaling a different axis than the parameter count the whole scaling era optimised.

The turn's public moment came in early 2025, when the Chinese lab DeepSeek released models at or near frontier quality reportedly trained at a small fraction of assumed frontier cost, using aggressive efficiency techniques of exactly the kinds above. Markets convulsed; the assumption that capability was strictly proportional to spending, the scaling era's core economic belief, visibly cracked in a week. However the details settle historically, the DeepSeek moment made "efficiency" the industry's organising word, and it is the reason your final project (Week 12) asks you to evaluate a lean or scale-free approach against a transformer baseline: the module is training you for the turn, not the era it is leaving.

13.4 The thesis, stated in full

KEY IDEA

The module thesis. The current wave succeeded by scaling: data, compute, and architecture multiplied together (Chapter 10′s ingredients, industrialised by Chapter 12′s laws). That recipe genuinely worked, and is now straining against cost, energy, data, and diminishing returns. The frontier is therefore turning in two complementary directions: **toward science**, building known structure into models so they need less brute force, and **toward lean**, extracting more capability per parameter, per joule, and per euro. The two turns converge on the same destination, capable models that are cheap, private, efficient, and physically sensible, and every remaining week of this module sits somewhere on that map. Your standing question for twelve weeks, for every technique and every headline: **is this more brute force, more science, or more lean?**

13.5 A third winter?

The question the whole booklet has been building toward, answered with the calibration it has been teaching. History's honest testimony comes in three parts. Hype cycles are real, the present resembles one, and winters arrive when promises outrun results, and current promises are extravagant by any historical standard. Winters punish promises, not science: the underlying techniques survived both previous collapses and compounded straight through them, so a funding winter, if one comes, would slow but not erase. And this cycle differs from both predecessors in one measurable way: massive deployed revenue and daily use, which 1974 and 1987 entirely lacked; busts of that kind (the railways, the dot-coms) historically destroy investors rather than technologies, leaving infrastructure behind.

So the defensible position, and the one this module asks you to be able to **argue** rather than merely hold: certainty in either direction is the error. Anyone assuring you that exponential progress continues indefinitely, or that collapse is imminent, is selling something, and Chapter 14 will give you the tools to inspect the sales pitch. You are studying the field mid-cycle, which is uncomfortable and is also the best possible seat: by Week 12, having seen the science-grounded and lean turns up close, you will write your own forecast with instruments, not vibes. That forecast, in a real sense, is the final exam of Week 1, sat eleven weeks late.

QUIZ WATCH · WEEK 6

Be able to: name the four walls with one sentence of mechanism each; describe both turns with two example techniques per turn and the wall each turn primarily attacks; state the module thesis in your own words; and give the three-part calibrated answer on a third winter (cycle is real; winters punish promises, not science; this cycle has revenue the others lacked).

14 A Field Guide to Reading AI News

COVERS Everything before, operationalised: a checklist for separating substance from hype in any AI announcement

TIME About 15 minutes, then use it forever

IN BRIEF Seventy-five years of history compress into seven questions. Apply them to any AI headline and you are doing what LO1 actually certifies: tracing development in order to **evaluate the present**.

History you cannot use is trivia. This chapter converts the booklet into an instrument: seven questions to put to any AI announcement, each anchored to an episode you now know. Bring this checklist to Week 11, where evaluating "substance versus hype" in emerging trends is an explicit lecture topic, and to the critical brief of your final project, where it is close to a mark scheme.

1. **What did the system actually do, and in what world?** Separate the demonstrated capability from the language around it, then ask whether the world was a swimming pool or the ocean (Chapter 5): fixed rules and clean inputs, or open-ended mess? Shakey was an "electronic person" in a prepared room; SHRDLU understood **blocks**. Most demos live in pools; most deployment is ocean.
2. **Would this survive a benchmark a rival can run?** The statistical turn's core hygiene (Chapter 9): claims ride numbers on shared tests, or they ride adjectives. AlexNet ended arguments in one afternoon **because** ImageNet was public and scored (Chapter 10). Beware equally the inverse failure, benchmarks gamed into meaninglessness; the question is whether **independent** evaluation exists, not whether a leaderboard was screenshotted.
3. **Whose supply made this possible, and is it durable?** The three-ingredient discipline (Chapter 10): breakthroughs ride supplies of data, compute, and architecture, often built by adjacent industries. Ask which supply moved, and whether it is compounding (GPU curves) or depleting (human text, Chapter 13). Claims that assume a depleting supply will keep flowing are 1985′s expert systems, priced before maintenance.
4. **Is the impressive part the engine or the interface?** ChatGPT's lesson (Chapter 12), and the perceptron press conference's (Chapter 5): adoption, packaging, and fluency each amplify perceived capability. Neither answer is damning; interfaces are real inventions. But know which you are admiring, and remember the ELIZA effect is now a design pattern, sometimes deliberately deployed.
5. **Who is speaking, and what are they selling?** Rosenblatt had Navy funding to renew; the Fifth Generation had a nation's prestige; today's speakers may be raising capital, recruiting, lobbying, or, subtler, seeking your attention with doom, which sells as well as utopia. Incentive analysis is not cynicism; it is source criticism, the same skill your other modules call information literacy.
6. **Which version of the idea does this criticism reach?** The XOR discipline (Chapter 6), for negative headlines: is the demonstrated limit structural to the whole approach, or a property of the simplest current version with an acknowledged escape route? The bicycle is not the wheel. Equally, "escape route exists" is not "escape route is easy": seventeen years passed between the XOR critique and backpropagation.
7. **Where does it sit on the module map?** The thesis question (Chapter 13), for any new technique: more brute force, more science, or more lean? By Week 12 you should place any announcement on that map in one sentence, which is the module's definition of understanding the field's present.

TRY IT YOURSELF

Standing exercise for the semester, and reliable forum gold: each week, take one AI headline through all seven questions in under two hundred words. By Week 6 this is quiz preparation; by Week 12 it is a draft paragraph of your final project's critical brief.

15 The Revision Core: The Whole Story on Two Pages

If you memorise nothing else for Week 6, memorise this chapter.

15.1 The cycle

Breakthrough → hype → promises outrun results → funding collapse (winter) → quiet progress under other names → next breakthrough. Two completed turns; currently high on the third rise; whether it ends in winter, plateau, or something new is open, and arguable either way with the Chapter 12 evidence.

15.2 The master timeline

Years	Era	Driving idea	Why it rose, and why it fell or turned
1650-1949	Prehistory	Reasoning as computation	Boole (logic as algebra), Shannon (algebra as circuits), Turing 1936 (universal machine), McCulloch-Pitts (neurons as logic): the inference "brains compute, so machines can" becomes available.
1950-1956	Founding	Behavioural test; a named field	Turing's Imitation Game makes intelligence measurable; Dartmouth names the field and overdoses on optimism; symbolic and connectionist families split.
1956-1969	Golden years	Symbolic search; first learning machine	Microworld triumphs (Logic Theorist, SHRDLU); perceptron learns from error; ELIZA exposes the fluency trap. Rises on ARPA money; falls to XOR critique and combinatorial explosion.
1974-1980	**Winter I**		ALPAC and Lighthill document the promise gap; funding collapses; neural research frozen by a correct critique of the simplest model, read too broadly.
1980-1987	Expert systems	Hand-written IF-THEN rules	Real commercial wins (XCON, MYCIN-class systems); rises on narrow, sellable promise; falls to brittleness, the knowledge bottleneck, and maintenance economics.
1987-1993	**Winter II**		Lisp machine market dies under commodity hardware; industry closes; "AI" becomes toxic; field rebrands as machine learning, which is also an honest signal of a method shift.
1990s-2011	Statistical turn	Learn patterns from data	ML colonises daily life (spam, credit, search, recommendations); probability replaces logic; benchmarks replace demos; Deep Blue is the old paradigm's

			finest hour and dead end; backprop and CNNs mature in the shadows.
2012-2017	Deep learning	Depth, at last affordable	AlexNet unites data (ImageNet), compute (gaming GPUs), architecture (CNN plus unblockers); method sweeps vision, speech, translation; AlphaGo's learned intuition beats the unsearchable game.
2017-2023	Scaling era	Attention; bigger is better, predictably	Transformer makes language learnable at GPU scale; scaling laws make progress purchasable; GPT-3′s emergence; RLHF plus a chat box detonate public adoption.
2023-now	The two turns	Past brute force	Four walls (cost, energy, data, returns); responses: science-grounded models (physics built in) and lean models (distillation, quantization, scale-free architectures, test-time reasoning); DeepSeek moment cracks the spend-equals-capability belief. Open ending: yours to forecast.

15.3 The three ingredients

Data plus compute plus architecture, **simultaneously**. Partial combinations and their eras: architecture alone is the perceptron (starved); compute alone is Deep Blue (no transfer); data alone is the 2000s web (shallow ceiling); data plus compute is ImageNet 2010 (arena, no champion); all three is AlexNet 2012. Corollary: stalled ideas may be waiting for a supply; ask which.

15.4 Eight one-line lessons the field paid for

1. Fluency is not understanding, and we are wired to confuse them (ELIZA, 1966; every chatbot since).
2. A correct criticism of the simplest version is not a refutation of the idea (XOR, 1969; the bicycle and the wheel).
3. Exponential spaces do not yield to brute force; "does not scale" is a structural verdict, not a complaint (combinatorial explosion, winter I).
4. Hand-written rules break outside anticipation, cannot capture tacit knowledge, and rot faster than repair, in open domains (the three diseases, 1987).
5. Winters punish promises, not science; a dormant field and a dead one look identical from outside (the trio through both winters).
6. Breakthroughs are often old ideas meeting new supply, frequently built by an adjacent industry for other reasons (AlexNet and the gamers, 2012).
7. The invention people adopt is often the interface, not the engine, and adoption has its own engineering and ethics (ChatGPT, 2022).
8. Every era's winning move meets a limit it cannot out-spend; the interesting question is always which wall, and which turn (this era: four walls, two turns).

QUIZ WATCH · WEEK 6

The Week 6 quiz draws its Week 1 items from exactly this chapter's territory: cycle-stage identification, timeline ordering, the three ingredients with era attributions, the winters' contrasting anatomies, and the lessons applied to short scenarios. If you can reconstruct this chapter from memory onto two blank pages, Week 1 revision is complete.

16 Glossary in Plain English

Terms are ordered by first appearance in the story rather than alphabetically, so the glossary doubles as a skeleton retelling. Each entry gives the plain meaning first, then the booklet chapter where it lives.

Term	Plain-English meaning	Ch.
Turing machine	Turing's 1936 thought experiment: an idealised device proving one flexible machine can run any program. The theoretical blueprint of every computer.	2
Lovelace's objection	The 1843 claim that machines only do what they are ordered to, so cannot originate. Machine learning complicates it; Turing rebutted it; AlphaGo's move 37 is its strongest counterexample.	2
McCulloch-Pitts neuron	The 1943 simplified brain cell: sum the inputs, fire over a threshold. Ancestor of every network unit, but with fixed, hand-set connections.	2
Turing Test	Judge a machine by whether its conversation is indistinguishable from a person's. Replaced "can machines think?" with something measurable, founding the field's benchmark culture.	3
Symbolic AI	The "write the rules down" family: intelligence as logic over symbols. Dominant 1956 to the 1990s; transparent, brittle.	4
Connectionism	The "learn from experience" family: networks of simple units that adjust themselves. Lost for thirty years, then won nearly everything.	4
Microworld	A radically simplified domain (blocks on a table) where methods shine because mess is excluded. The roped-off shallow end; its risk is extrapolating to the ocean.	5
Perceptron	Rosenblatt's 1958 learning machine: weighted evidence, a threshold, and a rule that nudges the weights when wrong. The darts loop; ancestor of all modern training.	5
Weight	A dial inside a network setting how strongly one input pushes the verdict. Learning is adjusting dials from error; modern models have billions.	5
Convergence theorem	Rosenblatt's guarantee: if the examples **can** be separated by his machine, training will find a separation. The fine print became the XOR crisis.	5
ELIZA effect	Our reflex to project understanding onto fluent language. A fact about humans, so it compounds as machines improve; now a design responsibility.	5
XOR	One or the other but not both (the two-switch stairwell light). Famous because one straight line cannot separate its cases, so single-layer perceptrons provably fail on it.	6
Credit assignment problem	When a many-layered system errs, which internal dial deserves blame? Unsolved in 1969; solved practically by backpropagation.	6
Backpropagation	The 1986-popularised bookkeeping that assigns each weight its share of an error, making deep networks trainable. The escape route from XOR, seventeen years late. Week 3's core topic.	6, 8

Combinatorial explosion	Possibility counts that multiply with every added element, producing exponential spaces no brute force can search. Winter I's honest technical cause.	6
AI winter	A collapse of funding and credibility after promises outrun results. Two so far (mid-70s, late-80s); winters punish promises, not science.	6, 8
Expert system	1980s product: thousands of hand-written IF-THEN rules plus an inference engine that chains them. Transparent, commercially real, and fatally rule-bound.	7
Inference engine	The generic machinery that fires matching rules, adds conclusions, and repeats. Separable from the knowledge base, which was the era's proudest design idea.	7
Brittleness	Failing absurdly and confidently just outside anticipated cases, with no sense of one's own limits. The phrasebook tourist; disease one of rules.	7
Tacit knowledge	What experts know but cannot tell (grandmother's salt). Uninterviewable, hence the knowledge bottleneck; learnable from examples, hence machine learning's wager.	7
Machine learning	Getting behaviour from labelled examples rather than hand-written rules. Partly a post-winter rebrand, wholly a method shift.	8
Benchmark culture	Claims ride numbers on shared public tests that rivals can run. The statistical turn's hygiene; the stage machinery of 2012.	9
AI effect	Redefining each achieved milestone as "not really intelligence". Contains real signal, but applied relentlessly it makes "real AI" unfalsifiable.	9
ImageNet	Fourteen million labelled photos and an annual contest: the data supply and the arena. Built against fashionable advice by Fei-Fei Li.	10
GPU	Graphics chip built for games: ten thousand adequate workers doing simultaneous small arithmetic, which is exactly what training is. The adjacent industry subsidy.	10
AlexNet	The 2012 network that united the three ingredients and demolished ImageNet by ten points, converting the field within a cycle. Mostly old ideas, newly affordable.	10
Three-ingredient thesis	Deep learning ignites only when big labelled data, cheap parallel compute, and trainable deep architecture coexist. Week 1′s single most important idea.	10
GAN	Two networks trained adversarially, forger against detective, yielding photorealistic generation. Opened the modern provenance and deepfake ethics agenda.	11
Self-play	Generating training experience by playing against copies of yourself; how AlphaGo surpassed its human teachers.	11
Transformer	The 2017 architecture whose attention mechanism relates every word to every other simultaneously; parallel-friendly, scale-friendly, and the basis of all current large language models.	12
Attention	The learned weighting of which other words matter for interpreting each word: the whole book open on the desk. Its cost grows with the square of length, motivating Week 8.	12

Large language model (LLM)	A huge transformer trained to predict the next word over internet-scale text, from which broader abilities emerge.	12
Scaling laws	The finding that capability improves smoothly and predictably with joint growth in model, data, and compute; made progress purchasable and financed the arms race.	12
Emergence	Abilities appearing at scale without being individually engineered (in-context learning). Real phenomenon, contested interpretation; handle with tongs.	12
RLHF	Reinforcement learning from human feedback: tuning a model against learned human preferences. The step that made a text predictor into an assistant; cybernetics' homecoming.	12
Distillation	Training a small student model to imitate a large teacher: capability at a fraction of the size. Lean turn, Week 9.	13
Quantization	Storing weights at lower numeric precision (four bits or fewer) to shrink and speed models for modest accuracy cost. Lean turn, Week 9.	13
LoRA	Parameter-efficient fine-tuning: adapt a huge model by training only a small attachment. Customisation on a laptop. Week 9.	13
State-space models	Scale-free architectures (Mamba and relatives) dodging attention's quadratic bill with linear-time, constant-memory designs. Week 8.	13
Reasoning models	Models that improve by spending more computation at answer time, scaling thinking rather than parameters.	13
Physics-informed networks	Models with governing equations built into training, cutting data needs by refusing to violate known science. Science turn, Week 5.	13
The module thesis	Scaling worked; it is hitting walls of cost, energy, data, and returns; the frontier turns toward science-grounded and lean models. The standing question: brute force, science, or lean?	13

17 Question Bank: Practising for Week 6

The Week 6 quiz mixes auto-graded items with short structured responses. Everything below is written in those registers. Answers are deliberately not printed; every question is answerable from a named chapter, and hunting the answer is the revision.

17.1 Short answer: two to three sentences each

1. State what the Turing Test replaced, and why replacing it was a **practical** act rather than a philosophical one. (Chapter 3)
2. Explain Lovelace's objection and give one modern development that complicates it. (Chapters 2, 11)
3. Contrast symbolic AI and connectionism in two sentences, naming one strength and one weakness of each. (Chapter 4)
4. Walk through the perceptron's learning loop in four plain-English steps, and name the Week 3 techniques that are this loop made precise. (Chapter 5)
5. What does the perceptron convergence theorem guarantee, and what is in its fine print? (Chapters 5, 6)
6. Define the ELIZA effect and explain why it grows stronger, not weaker, as machines improve. (Chapter 5)
7. Using the stairwell light, explain what XOR computes and why one straight line cannot separate its cases. (Chapter 6)
8. The XOR episode is often called an overreaction. Give the two facts that justify that description. (Chapter 6)
9. Define combinatorial explosion and explain why it is a wall rather than a difficulty. (Chapter 6)
10. Name the three diseases of expert systems and attach the booklet's analogy to each. (Chapter 7)
11. Which disease does machine learning specifically cure, and by what mechanism? (Chapter 7)
12. MYCIN matched specialists and was never deployed. What does the gap between those two facts teach, and which week of this module inherits the lesson? (Chapter 7)
13. Contrast the two winters: primary trigger, who administered the punishment, and what survived each. (Chapters 6, 8)
14. The post-winter rebrand was both camouflage and honest signal. Explain both halves. (Chapter 8)
15. Why is Deep Blue best classified with the old paradigm despite arriving in 1997? (Chapter 9)
16. Define the AI effect and explain the calibration trap on **each** side of it. (Chapter 9)
17. Name the three ingredients and attach a stalled historical era to any two partial combinations. (Chapter 10)
18. Explain the adjacent industry subsidy using the GPU story, and give the module's forward-looking use of the pattern. (Chapter 10)
19. Defend in three sentences: "2012 was a coincidence of supply, not a new idea." (Chapter 10)
20. Contrast Deep Blue and AlphaGo on method, role of human knowledge, and transferability. (Chapter 11)
21. Connect AlphaGo's move 37 to Lovelace's objection in no more than two sentences. (Chapters 2, 11)
22. Explain attention with the mystery-novel analogy, then name the cost the analogy hides and the week that addresses it. (Chapter 12)
23. What did scaling laws show, and what was the economic consequence of **predictability** specifically? (Chapter 12)
24. Define emergence with the required epistemic caution. (Chapter 12)
25. ChatGPT's engine predated it by two years. Name the two additions that made the product, and the general lesson. (Chapter 12)
26. Name the four walls of the scaling recipe with one sentence of mechanism each. (Chapter 13)
27. Describe the two turns, with two example techniques each, and the wall each turn primarily attacks. (Chapter 13)
28. Give the three-part calibrated answer to "will there be a third winter?". (Chapter 13)

17.2 Pattern identification: name the stage, justify in two sentences

For each scenario (all invented), identify the stage of the AI cycle it most resembles and the historical episode it echoes.

1. A startup demonstrates a warehouse robot performing flawlessly in a purpose-built demonstration facility, and its founder tells investors that fully general household robots are three years away.
2. A national newspaper reports that a new model "has taught itself chemistry", based on a benchmark the model's own developers designed and administered.
3. Following two high-profile product failures, a government research council announces it will fund "data science" and "decision systems" but pauses its "artificial intelligence" programme.
4. A company maintains twelve thousand hand-written business rules; each quarterly regulation change now requires a specialist team, and errors from rule interactions are rising.
5. A once-dismissed technique suddenly produces state-of-the-art results, and analysis shows the decisive change was a new public dataset ten times larger than any predecessor.
6. A firm's stock falls sharply after a rival demonstrates comparable model quality at one tenth the training cost.

17.3 Concept application: one paragraph each

1. A stakeholder asks: "Why can't we just write rules for detecting fraudulent insurance claims? Our senior assessors know fraud when they see it." Draft your reply using the three diseases, and state the one condition under which rules **would** be the right answer.
2. Take any real AI headline from the past month and run it through any four of Chapter 14′s seven questions.
3. Your colleague argues: "Large language models are just autocomplete; the criticism ends the discussion." Respond using the XOR discipline, and then state what the criticism nonetheless gets right, using the fluency trap.
4. Using the three-ingredient thesis, identify which single ingredient you believe is currently the binding constraint on frontier progress, and defend the choice against one alternative.
5. Place each of the following on the module map (brute force, science, or lean), with one justifying sentence each: a 4-bit quantized on-device assistant; a training run ten times larger than any predecessor; a climate model with atmospheric physics built into its loss; a system that answers hard questions by generating and checking many candidate chains of reasoning.

17.4 Discussion prompts: for the forum and the reading week

1. The field was named at a press-shy workshop and renamed by a funding crisis. How much does what we **call** this technology shape who funds it, who fears it, and who is accountable for it? Would "machine learning" have attracted the EU AI Act?
2. Weizenbaum concluded from ELIZA that some things machines could do, they should not do. Name one current application you would place in that category, and defend the placement against the strongest reply you can construct.
3. Fei-Fei Li was advised that building a dataset was service work, not science. In your intended career, what is the equivalent unglamorous infrastructure, and who is building it?
4. You have read eight lessons the field paid for (Chapter 15). Which one do you predict the industry is currently in the process of re-learning the hard way? Commit to a position; revisit it in Week 12.

18 Further Reading and Viewing

Everything here is optional enrichment; nothing in this chapter is examinable. Items are ordered by effort, not importance.

Under one hour. Turing's 1950 paper, **Computing Machinery and Intelligence**, remains the single best primary source in the field and is genuinely readable; at minimum, read the objections and replies. Fei-Fei Li's TED talk on teaching computers to see tells the ImageNet story from inside, including the unfashionable-bet years. The televised 1973 Lighthill debate (search "Lighthill debate") is winter I preserved in amber: watch five minutes to feel how a field sounds when it is defending its existence.

An evening. The documentary **AlphaGo** (2017) covers the Lee Sedol match from inside both camps, including move 37 and Lee's move 78; it is the best two hours of AI history on film and doubles as a study of the ELIZA effect, the AI effect, and machine originality in a single narrative. Chapter 1 of Russell and Norvig, **Artificial Intelligence: A Modern Approach** (2020), is the standard scholarly account of the history this booklet has told informally, and the module's core text; reading it after this booklet is the right order.

A weekend or more. Cade Metz, **Genius Makers** (2021), is a journalist's account of the connectionists' wilderness years and triumph, strong on the human texture of Chapters 8 to 10. Melanie Mitchell, **Artificial Intelligence: A Guide for Thinking Humans** (2019), is the best single book on calibration, sceptical and generous at once, and pairs naturally with Chapter 13′s field guide. Brian Christian, **The Alignment Problem** (2020), bridges this week's history to Week 2′s ethics agenda through the RLHF lineage. Weizenbaum's **Computer Power and Human Reason** (1976) is the field's first inside critique and startlingly current on the ELIZA effect at scale.

Primary papers, for the ambitious. McCulloch and Pitts (1943) for the founding neuron; Rosenblatt (1958) for the perceptron; Rumelhart, Hinton and Williams (1986) for backpropagation; Krizhevsky, Sutskever and Hinton (2012) for AlexNet; Vaswani and colleagues, **Attention Is All You Need** (2017), which you will revisit properly in Week 7, and which is worth skimming now purely to see what a field-defining paper looks like: eleven pages, one diagram everyone has since redrawn, and a title that undersold the next decade.

Appendix: Setting Up Your Environment

COVERS	Python, virtual environments, Git and GitHub, Google Colab, HuggingFace and Ollama
TIME	About 45 minutes, once
IN BRIEF	The lab's practical half. Nothing here is examinable, and everything here is required: by Week 3 you cannot follow the module without it. The condensed version lives in `week1-lab/INSTALL.md`.

Five tools carry the rest of this module. They do not all have the same deadline, and installing them all in one evening is a good way to run out of disk space five minutes before a lab.

Tool	What it is for	First needed
Python 3.11+	The language everything else runs on.	Week 1
A virtual environment	Keeps this module's packages away from everything else on your machine.	Week 1
Git and GitHub	Where every deliverable lives, including the Week 12 project.	Week 1
Google Colab	A free GPU for when your own machine is not enough.	Weeks 7–9
HuggingFace	Where the models and datasets come from.	Week 3
Ollama	Runs models entirely on your own machine, with no API key.	Weeks 7, 9

Only the first three are needed to leave the first lab in working order.

18.1 Python

You need **3.11 or newer**. Check with `python3 --version` (macOS, Linux) or `python --version` (Windows). If that fails, or reports 3.10 or older:

- **Windows.** Install from the Microsoft Store, or from python.org. On the python.org installer, **tick "Add python.exe to PATH"** on the first screen. Missing that box is the single most common cause of `python is not recognised`.
- **macOS.** `brew install python@3.12`, or the python.org installer. Do not rely on the `/usr/bin/python3` that Apple ships; it exists for the operating system's own use and is not yours to modify.
- **Linux.** `sudo apt install python3 python3-venv python3-pip` on Debian or Ubuntu. Note that `python3-venv` is a separate package and is easy to miss.

> **KEY IDEA**
>
> **Never install packages into your system Python.** Always work inside a virtual environment. Two modules will eventually want different versions of the same library, and the virtual environment is the thing that stops them fighting. If you ever see the error `externally-managed-environment`, that is not a bug: it is your operating system enforcing exactly this rule on your behalf.

18.2 The virtual environment

From inside the `week1-lab` folder:

```
python3 -m venv .venv           # create it, once
```

```
source .venv/bin/activate          # activate: macOS / Linux
.venv\Scripts\activate             # activate: Windows PowerShell

pip install -r requirements.txt    # the core packages: small and fast
```

Your prompt now begins with `(.venv)`. That prefix is the entire point: it tells you which Python you are about to run. When you open a new terminal it will be gone, and you activate again. Nothing has broken. To leave deliberately, type `deactivate`.

The lab folder ships a one-command version, `setup.sh` for macOS and Linux and `setup.ps1` for Windows, which creates the environment, installs the core packages and runs the environment check. The requirements are deliberately split in two: `requirements.txt` holds only what Week 1 needs and installs in under a minute, while `requirements-full.txt` adds the PyTorch stack for Week 3 onward and is roughly two gigabytes. Install the second one when you have the time and the disk, using `bash setup.sh --full`.

TRY IT YOURSELF

Run `python setup_check.py`. It never crashes, it installs nothing, and it prints one line per tool ending in a count of **blocking** items. Aim for zero blocking. The "sort out later" items have deadlines in Weeks 3 and 7. If something is stuck, bring that output to the lab: it tells the demonstrator in five seconds what would otherwise take five minutes of questions.

18.3 Git and GitHub

Your GitHub repository is not a formality: it is the module deliverable. Every week adds to it and the Week 12 project is submitted from it.

```
git config --global user.name  "Your Name"
git config --global user.email "you@setu.ie"

git init
git add .
git commit -m "Week 1: environment and timeline lab"
git branch -M main
git remote add origin https://github.com/YOURNAME/ai-evolution-2026.git
git push -u origin main
```

GitHub stopped accepting account passwords over HTTPS in 2021. When git asks for a password it wants a **personal access token** (Settings, then Developer settings, then Tokens). The alternatives are the GitHub CLI, `gh auth login`, which handles this for you, or an SSH key created with `ssh-keygen -t ed25519 -C "you@setu.ie"` and pasted into Settings.

A `.gitignore` ships in the lab folder. It keeps the virtual environment, caches, model weights and any `.env` file out of your repository. **Never commit a token.** Week 2 spends two hours on the proposition that what a system records is a decision with consequences; your repository is the first place you make it.

18.4 Google Colab

Colab is a notebook in a browser tab with a free GPU attached, at `colab.research.google.com`. Nothing to install. Choose Runtime, then Change runtime type, then a GPU, to get an accelerator.

To run this lab there, put the following in the first cell:

```
!git clone https://github.com/YOURNAME/ai-evolution-2026.git
%cd ai-evolution-2026/week1-lab
!pip install -q -r requirements.txt
```

Two things to know. Colab **forgets everything** when it disconnects, so anything you want to keep is pushed to GitHub or saved to Drive before you close the tab. And it is an escape hatch rather than a default: use it when your own machine cannot cope, realistically Weeks 7 to 9. For Weeks 1 to 5 a laptop is entirely sufficient, and working locally teaches you considerably more about what is actually happening.

18.5 HuggingFace

The Hub supplies the models and datasets of Weeks 3 to 11. Create a free account at huggingface.co, then Settings, then Access Tokens, then a new token with **read** access, which is all you need for now.

```
pip install -U huggingface_hub
hf auth login              # paste the token when prompted
hf auth whoami             # confirms who you are
```

DEEPER DETAIL: The CLI changed its name, and half the internet has not noticed

The command-line tool was called `huggingface-cli` until 2025, when it was renamed to `hf` and reorganised into a `hf <resource> <action>` grammar: `hf auth login`, `hf download`, `hf cache scan`, `hf repo create`. The old name still works and prints a deprecation warning pointing at the new one.

This matters practically, because a large fraction of the tutorials, Stack Overflow answers and blog posts you will find still use the old form. They are not wrong, merely dated. `huggingface-cli login` and `hf auth login` do the same thing. Recognising a renamed tool rather than concluding that a tutorial is broken is a small skill that will save you hours across this programme.

Downloads land in `~/.cache/huggingface` by default and grow quickly. `hf cache scan` shows what is there and `hf cache delete` reclaims the space; setting the `HF_HOME` environment variable moves the whole cache to another drive if your laptop is tight. Some models, Llama among them, are **gated**: you must accept a licence on the model page while logged in before the download will work. You will meet that in Week 7, not before.

18.6 Ollama

Ollama runs models entirely on your own machine. No API key, no account, and no data leaving the laptop. It is the runtime behind Weeks 7 and 9.

On macOS and Windows, download the installer from ollama.com. On Linux:

```
curl -fsSL https://ollama.com/install.sh | sh

ollama serve                    # start the background service
ollama pull llama3.2:1b         # about 1.3 GB: start small
ollama run  llama3.2:1b "In two sentences, what was the AI winter of the 1970s?"
```

It also exposes an HTTP API on `localhost:11434`, which is what `hello_models.py` uses and what your Week 10 retrieval application will talk to:

```
curl http://localhost:11434/api/generate -d '{
  "model": "llama3.2:1b", "prompt": "Why did AlexNet matter?", "stream": false
}'
```

As a rough guide to sizing, a one-billion-parameter model wants about two gigabytes of memory, a seven- or eight-billion model about eight, and anything larger wants a GPU. Start at `1b` and grow only if it is comfortable.

KEY IDEA

Do this one before Week 7 even if you do nothing else on this list. When a 1.3 GB file answers you, on your own laptop, with the wifi switched off, the lean turn of Chapter 13 stops being a claim on a slide and becomes something you have personally observed. It is the most persuasive five minutes in the module, and it is free.

18.7 When it goes wrong

Symptom	Cause	Fix
`python: command not found`	not on PATH, or it is `python3`	Try `python3`. On Windows, reinstall ticking “Add python.exe to PATH”.
`externally-managed-environment`	installing into system Python	Activate the virtual environment first. This error is the rule being enforced.
`(.venv)` vanished from the prompt	new terminal window	Activate again. Nothing is broken.
`ModuleNotFoundError` after installing	the editor is using a different interpreter	In VS Code: Command Palette, Python: Select Interpreter, choose `.venv/bin/python`.

`Support for password authentication was removed`	git wants a token, not a password	Personal access token, `gh auth login`, or an SSH key.
`401 Unauthorized` from the Hub	not logged in, or a gated model	`hf auth login`, and accept the licence on the model page.
Ollama: `connection refused` on 11434	the service is not running	`ollama serve` in a second terminal.
Colab: cannot connect to a GPU backend	free tier busy, or quota used	Wait, or continue on CPU. Weeks 1 to 5 do not need a GPU.

18.8 The checklist

Before the Week 2 lecture:

1. `python setup_check.py` reports zero blocking items.
2. A private GitHub repository exists and this folder is pushed to it.
3. `hf auth whoami` returns your username.
4. `ollama run llama3.2:1b "hello"` answers on your own machine.
5. You have opened one Colab notebook and watched a cell run.

Appendix: How Week 1 Maps to the Module

Module element	Week 1 connection
LO1 (trace historical development)	This entire booklet; assessed in the Week 6 quiz (50% of module mark). The operational meaning of "trace" is Chapter 14: use the history to evaluate the present.
Indicative content 1	History of AI/ML, pioneers, winters, resurgence: Chapters 2 to 13.
Indicative content 5	Influential breakthroughs and how they improved the state of the art: threaded throughout; focal treatments of the perceptron (Ch. 5), backpropagation (Ch. 6, 8), AlexNet (Ch. 10), AlphaGo (Ch. 11), the transformer and RLHF (Ch. 12).
Module thesis	Stated in full in Chapter 13; every subsequent week sits on its map (brute force, science, or lean).
Week 2 (Responsible AI)	Inherits: the ELIZA effect as design responsibility (Ch. 5), MYCIN's deployment gap as the sociotechnical lesson (Ch. 7), opacity and the Kasparov bug (Ch. 9), generation ethics (Ch. 11), adoption-scale responsibility (Ch. 12).
Week 3 (training)	Inherits: the darts loop made precise as gradient descent and backpropagation (Ch. 5, 6); PyTorch as the industrialised loop (Ch. 11).
Weeks 4-5 (architecture; science-grounded)	Inherits: architecture as assumption, from convolutions (Ch. 10) to physical law (Ch. 13, first turn).
Weeks 7-9 (LLMs; scale-free; lean)	Inherits: the transformer and its quadratic bill (Ch. 12), the four walls, and the lean toolkit (Ch. 13, second turn).
Weeks 10-12 (systems; trends; project)	Inherits: the interface lesson (Ch. 12), the field guide (Ch. 14) as the critical-brief mark scheme, and the third-winter question as your closing forecast (Ch. 13).
Interactive Companion	The hype-wave timeline is Chapters 2 to 13 as a scrubbable object; its three-ingredient experiment is Chapter 10 as a toy; its self-check quiz samples Chapter 17's registers.

End of the Week 1 Reference Booklet. Next: Week 2, Responsible AI, before we build anything, where the question stops being "what happened?" and becomes "who is accountable when it happens again?"